

CoMedBench: A Multi-Source Benchmark of Synthetic Medical Data Fidelity and Downstream Utility

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Abstract

Access to clinical data is essential for developing reliable healthcare machine learning systems, but direct use of electronic health records is constrained by privacy regulation, institutional review, data-use agreements, and the risk of re-identification. Synthetic data promises a practical alternative: it can preserve useful statistical and clinical structure while reducing exposure of sensitive patient records. Yet the evidence for synthetic data utility in healthcare remains fragmented. Prior studies often evaluate a single generator, one dataset, or a narrow downstream task, making it difficult to know when synthetic data can support model development and when it fails to preserve task-critical signal. We introduce **CoMedBench**, a reproducible benchmark that evaluates a family of generators under a common clinical-validity framework and one shared training and evaluation engine, spanning static tabular and temporal downstream tasks on established critical-care datasets. In total the benchmark spans 37 dataset-task pairs across two modalities consists of 20 static tabular and 17 temporal ICU time-series-drawn from seven public data sources: three intensive-care databases (**MIMIC-III**, **MIMIC-IV**, and **eICU**) together with the **UCI** Machine Learning Repository, the **CDC BRFS** diabetes cohort (2015), **NHANES** (1999-2014), and the **pycox** survival datasets (**GBSG** and **METABRIC**). The benchmark evaluates both statistical fidelity and task utility by comparing models trained and tested across real and synthetic data. In these settings, synthetic training data preserves most of the downstream signal: on tabular tasks the reference generator CoMed-CTGAN retains a mean AUROC utility (the synthetic-to-real performance ratio) of 90.6%, rising to 97.3% for the strongest generator, CoMed-TVAE. Temporal ICU tasks are harder and more generator-sensitive: CoMed-CTGAN retains 81.6% (AUROC) and only 64.0% under the imbalance-sensitive AUPRC, whereas CoMed-TVAE still retains ~95% (AUROC). By organizing evaluation around downstream clinical prediction rather than visual similarity alone, the benchmark clarifies where synthetic data is useful, where temporal structure remains a bottleneck, and how future generators should be assessed for healthcare AI.

Introduction

Modern healthcare AI depends on large, representative, and carefully curated clinical datasets (Habeheh and Gohel 2021). Electronic health records (EHRs), intensive-care databases, laboratory measurements, and longitudinal moni-

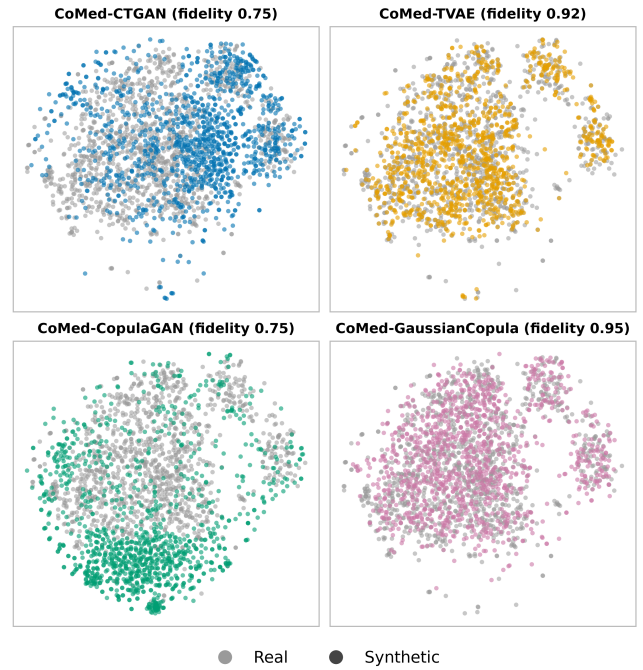


Figure 1: Two-dimensional t-SNE (perplexity 30, standardized features) of real vs. synthetic ICU records for the MIMIC-III heart-failure cohort. Generators differ markedly in how well synthetic data covers the real manifold: higher-fidelity models (CoMed-TVAE, CoMed-GC) mix throughout the real points, while lower-fidelity models (CoMed-CTGAN, CoMed-CopulaGAN) leave real regions uncovered. This gap between visual/statistical similarity and task usefulness motivates our benchmark.

toring streams contain signals that support mortality prediction, readmission modeling, length-of-stay estimation, and phenotyping. However, the same data are hard to share: clinical datasets include protected health information, rare diagnoses, and combinations of demographic and clinical attributes that can make patients identifiable even after conventional de-identification. As a result, many healthcare machine learning studies depend on a small number of public resources such as MIMIC-III, MIMIC-IV, and eICU, while

large institutional datasets remain inaccessible to most researchers (Johnson et al. 2016, 2023; Pollard et al. 2018).

Synthetic clinical data has emerged as a promising way to reduce this access barrier. Instead of releasing original patient records, a data holder trains a generative model and releases artificial records intended to approximate the distribution of the source data. If successful, synthetic data can enable algorithm prototyping, external benchmarking, software testing, and multi-institutional collaboration without exposing raw records.

The central challenge is that synthetic data must be useful, not merely realistic. A generator can match marginal distributions while distorting rare outcomes, comorbidity structure, medication-laboratory dependencies, or temporal dynamics. These distortions are harmful for downstream tasks because the predictive signal often lies in minority classes, rare events, and time-dependent trajectories (Pias et al. 2025). A synthetic dataset that appears faithful under aggregate statistics may therefore fail when used to train a model for a clinically relevant task; Figure 1 illustrates this gap.

Existing synthetic-data evaluations in healthcare report fidelity, privacy, and utility, but utility is often measured under different datasets, models, metrics, and task definitions, which makes comparison difficult. Researchers thus still lack a practical answer to a basic question: for which healthcare downstream tasks can synthetic data act as a useful substitute for real clinical data?

This paper addresses that gap with CoMedBench, a benchmark built to make synthetic-data evaluation controlled and comparable. CoMedBench applies a model-agnostic clinical-validity layer and wraps a family of generators in a single training pipeline so that the generator is the only component that varies between runs, and evaluates each synthetic table through two complementary scenarios: statistical fidelity, which measures how closely generated records resemble the source distribution, and downstream utility, which measures how well models trained on synthetic data transfer to real test data. This design separates apparent realism from practical usefulness.

Our work makes the following contributions:

- **Controlled evaluation under domain validity.** We evaluate four generator families under a common clinical-validity framework and one shared training and evaluation pipeline, so performance is compared without confounding differences in preprocessing, structural validation, or downstream evaluation.
- **A multi-source clinical benchmark.** 37 dataset-task pairs-20 static tabular and 17 temporal ICU time-series from seven public sources, each evaluated on statistical fidelity and downstream utility.
- **Synthetic data preserves downstream signal, but unevenly.** The strongest generator, CoMed-TVAE, retains $\sim 97\%$ of real-data AUROC on tabular tasks and $\sim 95\%$ on time-series; utility is high on static tabular tasks but degrades on temporal, imbalanced ICU tasks (AUPRC utility as low as $\sim 64\%$), and no single generator dominates across the evaluation settings.
- **Fidelity is a necessary but insufficient proxy.** Overall

statistical fidelity correlates with downstream utility only moderately (Overall $r = 0.67$, $\rho = 0.74$); utility must therefore be measured directly, with rare-event temporal ICU tasks the main bottleneck.

Related Works

Synthetic Data in Tabular and Time-Series Data

Synthetic data for structured data can be grouped into single-table, multi-table, and time-series synthesis. Single-table methods generate one flat table with mixed categorical and continuous columns, focusing on distributional fidelity, privacy, and machine-learning utility; examples include CTGAN-style models, differentially private normalizing flows, diffusion models, and autoencoder-based tabular generators (Xu et al. 2019; Patki, Wedge, and Veeramachaneni 2016; Lee et al. 2022). These methods suit static clinical cohorts but may lose key relationships when a relational EHR database is flattened into one table.

Multi-table synthesis instead preserves dependencies across interconnected tables (e.g., patient-admission, diagnosis-procedure, and laboratory-medication relationships), better reflecting real-world healthcare databases in which downstream labels depend on information distributed across multiple tables. Such relational methods are important for reconstructing realistic EHR structure rather than only generating independent rows.

Time-series synthesis focuses on sequential records where order, missingness, irregular sampling, and temporal dynamics are clinically meaningful. This direction is especially relevant for ICU tasks because prior work using models such as GRU-D and established MIMIC-III benchmarks has shown that temporal trends and missingness patterns are predictive for mortality, decompensation, length of stay, and phenotyping (Che et al. 2018; Harutyunyan et al. 2019).

Synthetic Data in Downstream Tasks

Synthetic data is most useful when it supports downstream tasks, not only when it looks statistically similar to real data. A common evaluation is train-on-synthetic, test-on-real (TSTR): models trained on generated data are evaluated on a held-out real test set, directly measuring whether synthetic data preserves transferable predictive signal.

Established MIMIC-III and MIMIC-IV benchmarks provide common structured and time-series tasks-mortality, readmission, length-of-stay, and phenotyping-for evaluating whether synthetic clinical data preserves useful predictive signal (Harutyunyan et al. 2019; Che et al. 2018; Gupta et al. 2022b,a; Bui, Warriar, and Gupta 2024; Li et al. 2021; Peng et al. 2022; Qiu et al. 2022; Lin et al. 2019; Adisa 2026; Kakadiaris 2023). These tasks motivate our benchmark because synthetic data may perform well on static tabular prediction while failing to preserve temporal patterns or rare-event signals.

Synthetic Data in Healthcare AI

Public datasets such as MIMIC-III, MIMIC-IV, and eICU have enabled critical-care research, but broad access to in-

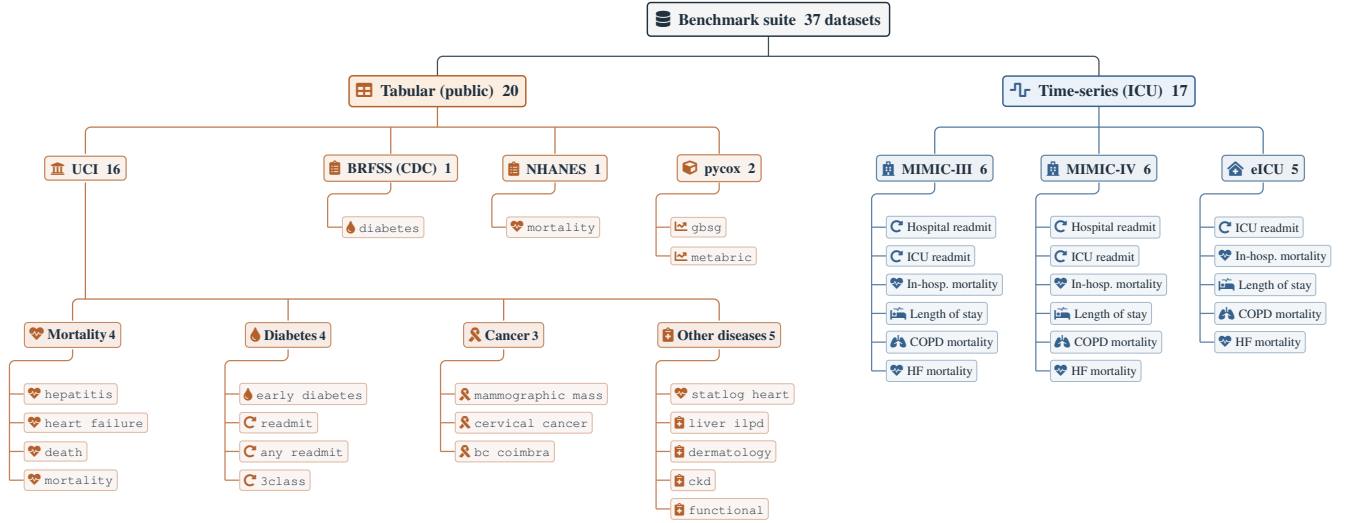


Figure 2: Benchmark coverage. The 37 dataset-task pairs split by modality (tabular/public vs. time-series/ICU) and by source; the public UCI datasets are further grouped into clinical subclasses (mortality, diabetes, cancer, and other diseases), while each ICU database (MIMIC-III, MIMIC-IV, eICU) contributes mortality, readmission, and length-of-stay tasks.

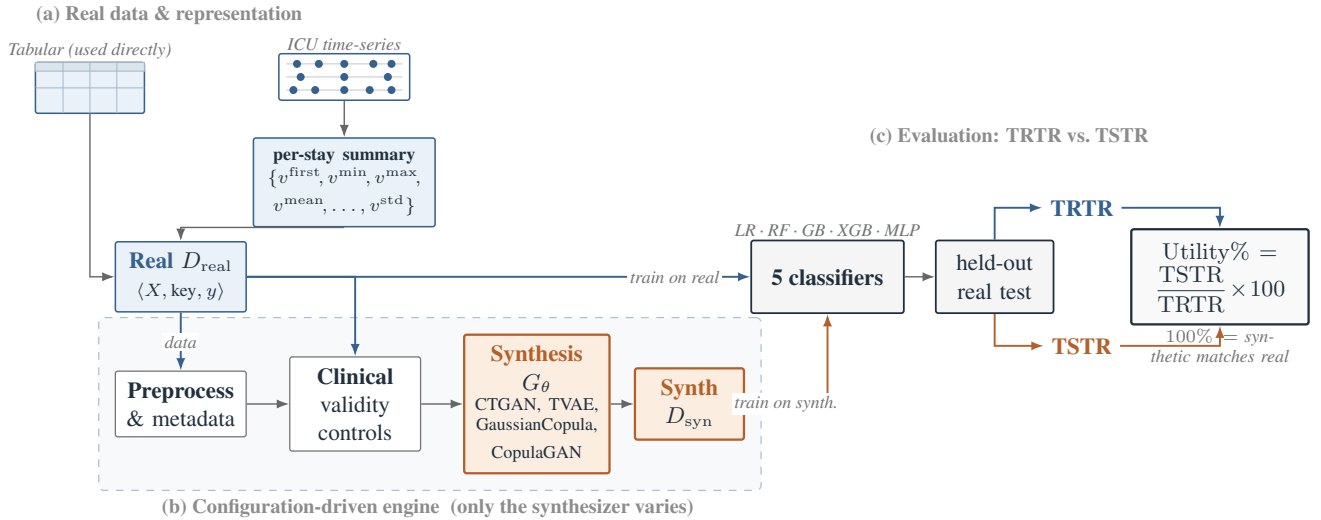


Figure 3: The CoMedBench pipeline. Real data (used directly as a flat table, or an ICU stay’s multivariate series summarized into a per-stay feature vector) drives a single configuration-driven engine—preprocessing and metadata, clinical-validity controls, and a synthesizer G_θ —whose only stage that changes across runs is the synthesizer. The resulting synthetic table D_{syn} and the real table D_{real} are then compared by training the same five classifiers on each and testing on an *identical* held-out real set (TRTR vs. TSTR), yielding the utility ratio.

stitutional EHR data remains limited (Johnson et al. 2016, 2023; Pollard et al. 2018).

Synthetic healthcare data can support prototyping, testing, education, and benchmarking without exposing raw patient records (Chen et al. 2021; Tucker et al. 2020). However, privacy protection alone is insufficient: generated data must also preserve clinical structure such as outcome prevalence, comorbidity patterns, medication-laboratory relationships, and temporal deterioration. This motivates our benchmark,

which evaluates whether synthetic data supports established downstream tasks rather than only matching broad statistical properties.

Methodology

Overview of the CoMedBench Framework

CoMedBench treats synthetic clinical-data generation and evaluation as a single, configuration-driven procedure ap-

plied uniformly to every (dataset $\times$ task) pair. Each experiment is defined by a compact task specification: the input feature matrix, the primary-key column, the prediction label and the feature-typing rule. One shared engine consumes this specification and executes four stages (Figure 3): (1) *preprocessing and metadata*, which loads the flat feature matrix, imputes missing values, and assigns each column an explicit type; (2) *clinical-validity controls*, a model-agnostic layer that enforces schema and task-specific validity rules consistently across every generator; (3) *synthesis*, which fits a generator G_θ and samples a synthetic table D_{syn} whose label distribution matches the real class proportions; and (4) *evaluation*, which scores D_{syn} against the real data D_{real} along two axes, statistical *fidelity* and downstream machine-learning *utility* (TRTR vs. TSTR). Because the same four stages are applied to every dataset, task, and generator, only the synthesis stage (or a single ablated setting) changes between runs, making the study a controlled benchmark rather than a comparison of loosely coupled implementations.

Datasets, Tasks, and Representation

We evaluate on three intensive-care databases - MIMIC-III, MIMIC-IV, and eICU (Johnson et al. 2016, 2023; Pollard et al. 2018), together with a collection of public clinical and tabular datasets (Strack et al. 2014; Patrício et al. 2018; Ramana et al. 2011; Chicco and Jurman 2020; Katzman et al. 2018; Xie et al. 2019; Pias et al. 2026). Tasks are binary or multi-class prediction: in-hospital mortality, ICU and hospital 30-day readmission, ICU length of stay, and disease/outcome classification. Figure 2 shows how the suite breaks down by modality, source, and task.

Every task is modeled as a single flat table (one row per patient or stay). In the *tabular* representation the dataset is already a flat feature matrix and is used directly. In the *time-series-to-tabular* representation, each ICU stay’s multivariate time series is summarized into a per-stay feature vector: each clinical variable v is represented by v_{first} , v_{min} , v_{max} , and v_{mean} (and, where available, v_{last} , v_{median} , v_{std}) alongside demographic and contextual features, yielding one fixed-width row per stay.

Preprocessing and Metadata

Each table is loaded and assigned a primary key, a native identifier where one exists (e.g., stay id) or a surrogate key otherwise; rows with a missing label are dropped. We build single-table metadata assigning every column an explicit type: the key as an identifier (regenerated at sampling time), the label as categorical, and each feature as categorical or numerical. Missing values follow a domain-aware imputation policy, and low-signal columns are pruned (Tang et al. 2020). High-cardinality categorical fields are encoded so that generators emit only categories observed in training.

Clinical Validity Controls

To ensure that generated records satisfy basic clinical and structural validity requirements, we apply a model-agnostic domain-validity layer consistently across all evaluated generators. The layer uses schema information and task-specific

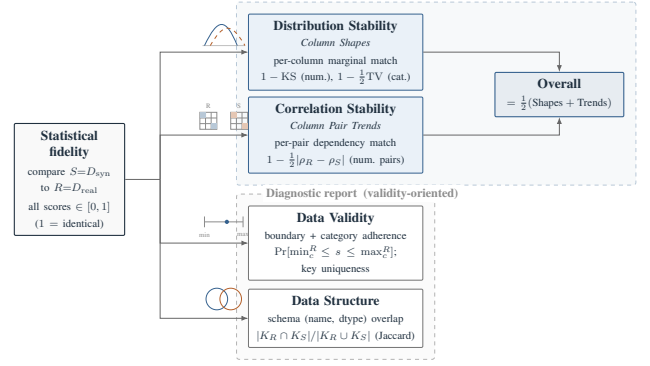


Figure 4: The CoMedBench fidelity suite. A *Quality* score combines Distribution Stability (per-column marginals) and Correlation Stability (column-pair trends); a *Diagnostic* score adds Data Validity and Data Structure. All scores lie in $[0, 1]$ (1 = identical).

validity rules to prevent clinically impossible or structurally invalid outputs. This study evaluates the effect of applying the same validity layer across generator families.

Synthesis: Generator Families

The same clinical-validity layer is applied identically to four single-table generators from distinct families, so the synthesis stage is the only component that changes between runs. **CoMed-CTGAN** is a conditional tabular generative adversarial network with mode-specific normalization and training-by-sampling for imbalanced discrete columns (Xu et al. 2019). **CoMed-TVAE** is a tabular variational autoencoder, **CoMed-CopulaGAN** a copula-augmented conditional GAN, and **CoMed-GaussianCopula** a statistical Gaussian-copula model with no training epochs. The GAN-based and TVAE-based generators use standard published configurations. Because the labels are imbalanced, generation is *label-conditioned*: the synthetic label distribution matches the real class proportions, and the number of synthetic rows equals the real dataset size.

Evaluation

We evaluate every synthetic table against the real table on two axes: statistical fidelity and downstream machine-learning utility (TRTR/TSTR). For brevity write $R = D_{\text{real}}$ and $S = D_{\text{syn}}$, and let $\mathcal{C}$ be the set of modeled columns, partitioned into numerical columns $\mathcal{C}_{\text{num}}$ and categorical columns $\mathcal{C}_{\text{cat}}$, with $\mathcal{P}$ the set of column pairs.

Statistical fidelity. Figure 4 organizes the fidelity metrics used below; each is computed by CoMedBench from standard column and column-pair statistics. *Distribution Stability* averages a per-column marginal score: the Kolmogorov-Smirnov complement for numerical columns and the total-variation complement for categorical columns. For a numerical column c with empirical CDFs $\hat{F}_c^R, \hat{F}_c^S$,

$$\text{KSC}(c) = 1 - \sup_x |\hat{F}_c^R(x) - \hat{F}_c^S(x)|, \quad (1)$$

and for a categorical column c with category set $\mathcal{A}_c$ and relative frequencies $R_c(a), S_c(a)$,

$$\text{TVC}(c) = 1 - \frac{1}{2} \sum_{a \in \mathcal{A}_c} |R_c(a) - S_c(a)|. \quad (2)$$

Distribution Stability is their mean over columns,

$$\text{Shapes} = \frac{1}{|\mathcal{C}|} \left(\sum_{c \in \mathcal{C}_{\text{num}}} \text{KSC}(c) + \sum_{c \in \mathcal{C}_{\text{cat}}} \text{TVC}(c) \right). \quad (3)$$

Correlation Stability averages a per-pair dependency score. For a numerical pair (a, b) with Pearson correlations ρ_{ab}^R, ρ_{ab}^S ,

$$\text{CS}(a, b) = 1 - \frac{|\rho_{ab}^R - \rho_{ab}^S|}{2}, \quad (4)$$

and for any pair involving a categorical column (numerical columns discretized into bins) with joint relative frequencies $R_{ab}(i, j), S_{ab}(i, j)$,

$$\text{CT}(a, b) = 1 - \frac{1}{2} \sum_{i, j} |R_{ab}(i, j) - S_{ab}(i, j)|. \quad (5)$$

Correlation Stability is

$$\text{Trends} = \frac{1}{|\mathcal{P}|} \sum_{(a, b) \in \mathcal{P}} m(a, b), \quad (6)$$

where

$$m(a, b) = \begin{cases} \text{CS}(a, b), & a, b \in \mathcal{C}_{\text{num}}, \\ \text{CT}(a, b), & \text{otherwise}, \end{cases} \quad (7)$$

and the *Overall* quality score is the mean of the two properties,

$$\text{Overall} = \frac{1}{2} (\text{Shapes} + \text{Trends}). \quad (8)$$

The diagnostic report additionally measures *Data Validity* and *Data Structure*. Data Validity averages a per-column adherence score: boundary adherence for numerical columns and category adherence for categorical columns,

$$\text{BA}(c) = \frac{1}{|S|} \sum_{s \in S_c} \mathbf{1}[\min_c R \leq s \leq \max_c R], \quad (9)$$

$$\text{CA}(c) = \frac{1}{|S|} \sum_{s \in S_c} \mathbf{1}[s \in \mathcal{A}_c^R], \quad (10)$$

with primary keys scored by uniqueness $\text{KU} = 1 - \frac{1}{|S|} \#\{\text{missing or duplicate synthetic keys}\}$. Data Structure is the Jaccard overlap of the (column name, dtype) schemas $\mathcal{K}^R, \mathcal{K}^S$,

$$\text{Structure} = \frac{|\mathcal{K}^R \cap \mathcal{K}^S|}{|\mathcal{K}^R \cup \mathcal{K}^S|}. \quad (11)$$

All fidelity scores lie in $[0, 1]$ ($1 = \text{identical}$).

Machine-learning utility (TRTR/TSTR). For each task we train five classifiers- Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and an MLP, under two regimes: *TRTR* (Train Real, Test Real; upper-bound reference) and *TSTR* (Train Synthetic, Test Real, on the *same* held-out real test set). We summarize the gap as a Utility ratio,

$$\text{Utility}\% = \frac{\text{AUC}_{\text{TSTR}}}{\text{AUC}_{\text{TRTR}}} \times 100, \quad (12)$$

so 100% means synthetic-trained models match real-trained ones.

Results

We report statistical fidelity across the four generators, downstream utility, and the relationship between them. Fidelity is measured for all four synthesizers, with a per-dataset breakdown in Table 2. Downstream utility is reported per dataset for all four synthesizers in Table 3 (AUROC and AUPRC), each aggregated over the five classifiers.

Statistical Fidelity

Table 2 reports per-dataset Overall fidelity for all four generators on both modalities. Averaging across datasets, CoMed-TVAE is the most faithful generator on both modalities, attaining the highest mean Overall score (0.902 tabular, 0.887 time-series) and winning on the majority of individual datasets. On tabular data the other three generators are closely grouped (mean Overall 0.85–0.87), with CoMed-GaussianCopula (0.867) slightly ahead of the GAN-based CoMed-CTGAN (0.860) and CoMed-CopulaGAN (0.848). On time-series data CoMed-GaussianCopula is the clear runner-up (mean 0.880), while CoMed-CTGAN (0.833) and CoMed-CopulaGAN (0.831) trail. Table 1 isolates the effect of the clinical-validity layer: it improves Overall fidelity for the GAN-based CoMed-CTGAN and CoMed-CopulaGAN on nearly every affected task (up to +0.098 on cervical cancer), while the CoMed-TVAE and CoMed-GaussianCopula models change only marginally.

Downstream Utility

Table 3 reports per-dataset downstream utility (AUROC and AUPRC) for all four synthesizers. On tabular tasks the reference generator CoMed-CTGAN preserves most of the real-data signal (mean AUC-ROC utility 90.6%, median 89.3%), and the strongest generator CoMed-TVAE reaches 97.3%; several datasets meet or exceed the real baseline (GBSG 100.5%, Mammographic Mass 100.0%). Time-series tasks are harder and more generator-sensitive: CoMed-CTGAN retains a mean AUC-ROC utility of 81.6%, whereas CoMed-TVAE still retains $\sim 95\%$. The gap widens sharply under AUC-PRC, which is sensitive to minority-class recovery: for CoMed-CTGAN, tabular utility averages 86.4% but time-series falls to 64.0%, with rare-outcome ICU mortality tasks retaining only $\sim 30\text{--}45\%$. Thus synthetic data transfers well for static tabular prediction, but the weaker generators lose substantial minority-class signal on temporal, highly imbalanced ICU tasks.

The distributions in tabular utility are higher and tighter (median near 93% across classifier-dataset pairs), while time-series utility is lower and more dispersed. The choice of downstream classifier matters relatively little, so the data modality, not the classifier, drives utility. Notably, on the imbalanced ICU tasks a synthetic dataset can retain acceptable AUROC while its AUPRC collapses, so AUROC alone overstates usefulness for rare-outcome prediction.

Does Fidelity Predict Utility?

Finally we relate the two axes across all 37 dataset-task pairs: higher fidelity is associated with higher utility, but the strength depends on the property. For the reference generator

Dataset	Task	CoMed-CTGAN	CoMed-TVAE	CoMed-CopulaGAN	CoMed-GaussianCopula
UCI Cervical Cancer	Cervical Cancer	0.867 (+0.098)	0.913 (−0.023)	0.828 (+0.044)	0.877 (+0.100)
MIMIC-IV	Heart Failure Mortality	0.825 (+0.038)	0.883 (−0.007)	0.822 (+0.030)	0.866 (−0.016)
MIMIC-III	Heart Failure Mortality	0.754 (+0.016)	0.915 (−0.006)	0.749 (+0.013)	0.955 (+0.000)
MIMIC-III	Length-of-Stay	0.840 (+0.006)	0.895 (−0.013)	0.841 (+0.010)	0.871 (−0.040)
MIMIC-IV	Length-of-Stay	0.916 (+0.004)	0.902 (−0.001)	0.913 (+0.003)	0.926 (+0.000)

Table 1: Effect of the clinical-validity layer on statistical fidelity. Each cell reports Overall fidelity with the validity layer, and in parentheses the change relative to the same generator without it (positive = the validity layer improves fidelity). The layer improves fidelity for the GAN-based CoMed-CTGAN and CoMed-CopulaGAN on nearly every task (up to +0.098), while the CoMed-TVAE and CoMed-GaussianCopula models change only marginally ($|\Delta| \leq 0.04$; slightly negative for CoMed-TVAE).

Tabular						Time-series					
Dataset	Task	CoMed-CTGAN	CoMed-TVAE	CoMed-CG	CoMed-GC	Dataset	Task	CoMed-CTGAN	CoMed-TVAE	CoMed-CG	CoMed-GC
Breast Cancer Coimbra	Breast Cancer	0.744	0.893	0.760	0.895	MIMIC-III	COPD Mortality	0.796	0.893	0.791	0.904
CDC BRFSS	Diabetes	0.970	0.940	0.940	0.849	MIMIC-III	Heart Failure Mortality	0.754	0.915	0.749	0.955
GBSG (pycox)	Breast Cancer Survival	0.935	0.967	0.909	0.883	MIMIC-III	Hospital Readmission	0.829	0.892	0.829	0.875
METABRIC (pycox)	Breast Cancer Survival	0.953	0.979	0.929	0.907	MIMIC-III	Length-of-Stay	0.840	0.895	0.841	0.871
SUPPORT2	Death	0.828	0.866	0.825	0.867	MIMIC-III	ICU Readmission	0.836	0.892	0.832	0.875
SUPPORT2	Functional Outcome	0.816	0.879	0.847	0.823	MIMIC-III	Hospital Mortality	0.846	0.873	0.847	0.815
SUPPORT2	Mortality	0.846	0.873	0.810	0.863	MIMIC-IV	COPD Mortality	0.805	0.885	0.810	0.858
UCI Cervical Cancer	Cervical Cancer	0.867	0.913	0.828	0.877	MIMIC-IV	Heart Failure Mortality	0.825	0.883	0.822	0.866
UCI Kidney Disease	Chronic Kidney Disease	0.846	0.870	0.838	0.819	MIMIC-IV	Hospital Readmission	0.845	0.928	0.887	0.915
UCI Dermatology	Dermatological Disease	0.905	0.966	0.899	0.841	MIMIC-IV	Length-of-Stay	0.916	0.902	0.913	0.926
UCI Diabetes	30-day Readmission	0.794	0.743	0.788	0.825	MIMIC-IV	ICU Mortality	0.856	0.890	0.857	0.845
UCI Diabetes	Any-Readmission	0.795	0.765	0.786	0.827	MIMIC-IV	ICU Readmission	0.754	0.894	0.749	0.912
UCI Diabetes	Readmission (3-class)	0.785	0.765	0.763	0.824	eICU	COPD Mortality	0.800	0.840	0.809	0.865
UCI Early Diabetes	Early Diabetes Risk	0.886	0.958	0.863	0.922	eICU	Heart Failure Mortality	0.832	0.860	0.819	0.872
UCI Heart Failure	Heart Failure Mortality	0.880	0.929	0.835	0.922	eICU	Length-of-Stay	0.857	0.863	0.865	0.863
UCI Hepatitis	Hepatitis Mortality	0.896	0.940	0.884	0.830	eICU	Mortality	0.907	0.884	0.915	0.860
UCI Indian Liver	Liver Disease	0.830	0.936	0.855	0.918	eICU	Readmission	0.863	0.890	0.854	0.885
UCI Mammographic	Mass Malignancy	0.951	0.977	0.940	0.839						
UCI Statlog Heart	Heart Disease	0.903	0.946	0.863	0.829						
NHANES	Mortality	0.767	0.944	0.804	0.944						

Table 2: Per-dataset Overall fidelity for the four generators on tabular (left) and time-series (right) tasks. Bold marks the best model per dataset.

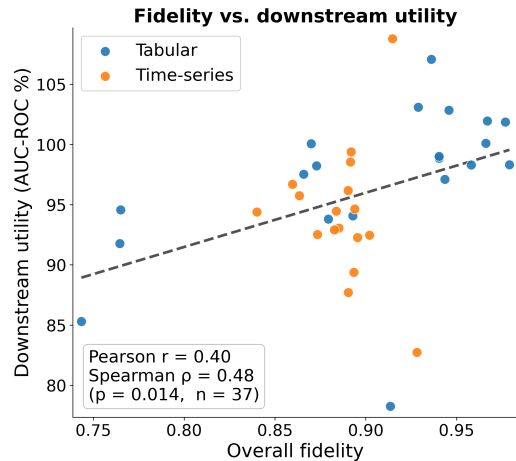


Figure 5: Overall fidelity vs. downstream utility (CoMed-TVAE AUC-ROC utility) across all 37 dataset-task pairs. Overall fidelity is the strongest single predictor of utility (Pearson $r = 0.40$, Spearman $\rho = 0.48$, $p = 0.014$); Correlation Stability is intermediate ($r = 0.33$) and marginal Distribution Stability weakest ($r = 0.25$), while dataset size is not predictive ($r = -0.27$, $p = 0.13$).

CoMed-CTGAN, Overall fidelity is the best single predictor ($r = 0.67$, $\rho = 0.74$), Correlation Stability is intermediate ($r = 0.58$), and marginal Distribution Stability is weakest ($r = 0.45$). Dataset size shows no significant relationship ($r = -0.20$, $p = 0.26$). The same ranking of predictors holds for the strongest generator, CoMed-TVAE (Figure 5), although the correlation is attenuated (Overall $r = 0.40$, $\rho = 0.48$) because its near-ceiling utility leaves little variance for fidelity to explain. These correlations are positive but imperfect, reinforcing the benchmark’s premise: fidelity is a useful but incomplete proxy, and downstream utility must be measured directly.

Conclusion

We presented CoMedBench, a reproducible benchmark that evaluates synthetic clinical data under a common clinical-validity framework along two axes—statistical fidelity and downstream task utility—using a single configuration-driven engine over three intensive-care databases and a suite of public clinical and tabular datasets. Across four generators, no single one dominates: CoMed-TVAE is the most faithful overall, while the Gaussian-copula model CoMed-GC is competitive or best on several small or highly imbalanced

Dataset	Task	Real	CoMed-CTGAN	CoMed-TVAE	CoMed-CopulaGAN	CoMed-GaussianCopula
<i>Tabular</i>						
BC Coimbra	Breast Cancer	0.777±0.124, 0.771±0.073	0.617±0.144, 0.671±0.094	0.731±0.066, 0.778±0.043	0.646±0.083, 0.752±0.051	0.710±0.109, 0.781±0.105
CDC BRFS	Diabetes	0.819±0.012, 0.401±0.030	0.800±0.015, 0.377±0.023	0.809±0.011, 0.382±0.022	0.755±0.020, 0.309±0.024	0.758±0.038, 0.323±0.045
GBSG	BC Survival	0.663±0.022, 0.691±0.026	0.666±0.022, 0.710±0.011	0.676±0.007, 0.714±0.009	0.654±0.015, 0.699±0.006	0.661±0.007, 0.707±0.010
METABRIC	BC Survival	0.721±0.014, 0.786±0.018	0.667±0.027, 0.724±0.038	0.708±0.034, 0.781±0.028	0.690±0.011, 0.759±0.013	0.696±0.042, 0.764±0.032
SUPPORT2	Death	0.848±0.005, 0.922±0.004	0.658±0.043, 0.795±0.028	0.827±0.005, 0.906±0.003	0.660±0.032, 0.802±0.025	0.764±0.008, 0.868±0.007
SUPPORT2	Mortality	0.949±0.002, 0.893±0.006	0.847±0.036, 0.665±0.051	0.933±0.005, 0.852±0.010	0.824±0.037, 0.648±0.071	0.824±0.085, 0.660±0.114
UCI Cervical Cancer	Cervical Cancer	0.603±0.108, 0.125±0.051	0.533±0.053, 0.087±0.014	0.472±0.043, 0.091±0.037	0.493±0.084, 0.106±0.030	0.496±0.059, 0.075±0.013
UCI CKD	CKD	0.999±0.002, 0.998±0.003	0.989±0.004, 0.978±0.009	0.999±0.001, 0.999±0.001	0.990±0.005, 0.981±0.013	0.960±0.036, 0.934±0.059
UCI Diabetes-130	30-day Readmit	0.666±0.009, 0.216±0.009	0.513±0.009, 0.114±0.002	0.568±0.027, 0.156±0.021	0.548±0.023, 0.132±0.011	0.593±0.028, 0.154±0.023
UCI Diabetes-130	Any Readmit	0.690±0.014, 0.653±0.013	0.614±0.008, 0.567±0.016	0.653±0.003, 0.609±0.012	0.611±0.013, 0.559±0.019	0.642±0.009, 0.601±0.016
UCI Early Diabetes	Early Diabetes	0.994±0.005, 0.997±0.003	0.988±0.002, 0.993±0.001	0.977±0.026, 0.987±0.015	0.963±0.006, 0.978±0.004	0.917±0.028, 0.953±0.016
UCI Heart Failure	HF Mortality	0.774±0.037, 0.599±0.062	0.691±0.047, 0.513±0.068	0.798±0.028, 0.693±0.071	0.768±0.028, 0.670±0.079	0.690±0.077, 0.571±0.076
UCI Hepatitis	Hepatitis Mort.	0.878±0.033, 0.972±0.009	0.867±0.050, 0.967±0.015	0.869±0.039, 0.969±0.009	0.702±0.155, 0.922±0.037	0.653±0.247, 0.886±0.104
UCI ILPD	Liver Disease	0.685±0.046, 0.436±0.075	0.669±0.028, 0.428±0.056	0.734±0.011, 0.472±0.013	0.603±0.032, 0.404±0.025	0.674±0.091, 0.435±0.087
UCI Mamm. Mass	Malignancy	0.886±0.016, 0.835±0.043	0.886±0.018, 0.860±0.024	0.902±0.008, 0.876±0.026	0.873±0.016, 0.834±0.030	0.700±0.096, 0.659±0.123
UCI Statlog Heart	Heart Disease	0.880±0.016, 0.842±0.033	0.852±0.032, 0.825±0.036	0.905±0.018, 0.888±0.024	0.846±0.022, 0.795±0.051	0.669±0.165, 0.639±0.147
NHANES	Mortality	0.920±0.005, 0.762±0.015	0.544±0.056, 0.224±0.038	0.893±0.029, 0.682±0.101	0.769±0.027, 0.409±0.040	0.868±0.049, 0.658±0.053
<i>Time-series</i>						
MIMIC-III	COPD Mort.	0.712±0.010, 0.343±0.027	0.412±0.026, 0.123±0.013	0.636±0.026, 0.227±0.026	0.445±0.045, 0.131±0.020	0.537±0.027, 0.190±0.027
MIMIC-III	HF Mortality	0.745±0.155, 0.413±0.170	0.480±0.109, 0.127±0.023	0.810±0.027, 0.453±0.030	0.461±0.059, 0.123±0.032	0.755±0.043, 0.336±0.045
MIMIC-III	Hosp. Readmit	0.600±0.017, 0.076±0.003	0.512±0.030, 0.058±0.006	0.596±0.022, 0.078±0.003	0.517±0.020, 0.057±0.005	0.513±0.054, 0.057±0.008
MIMIC-III	Length-of-Stay	0.757±0.014, 0.374±0.021	0.614±0.038, 0.238±0.019	0.699±0.024, 0.289±0.019	0.647±0.033, 0.256±0.016	0.612±0.063, 0.239±0.048
MIMIC-III	In-Hosp. Mortality	0.835±0.011, 0.467±0.019	0.690±0.019, 0.270±0.031	0.773±0.015, 0.358±0.015	0.665±0.015, 0.245±0.027	0.623±0.027, 0.210±0.027
MIMIC-III	ICU Readmit	0.601±0.016, 0.137±0.009	0.522±0.036, 0.107±0.013	0.592±0.014, 0.135±0.003	0.497±0.020, 0.099±0.005	0.512±0.036, 0.105±0.012
MIMIC-IV	COPD Mort.	0.827±0.014, 0.512±0.034	0.556±0.027, 0.220±0.031	0.770±0.006, 0.405±0.011	0.546±0.034, 0.228±0.032	0.557±0.074, 0.240±0.046
MIMIC-IV	HF Mortality	0.826±0.009, 0.474±0.013	0.555±0.061, 0.181±0.043	0.768±0.018, 0.372±0.016	0.632±0.058, 0.230±0.042	0.561±0.118, 0.209±0.083
MIMIC-IV	Hosp. Readmit	0.622±0.009, 0.359±0.014	0.519±0.009, 0.266±0.010	0.514±0.008, 0.269±0.012	0.502±0.011, 0.258±0.009	0.511±0.015, 0.265±0.011
MIMIC-IV	Length-of-Stay	0.828±0.008, 0.668±0.024	0.794±0.010, 0.606±0.017	0.766±0.013, 0.577±0.019	0.788±0.012, 0.605±0.018	0.750±0.026, 0.556±0.032
MIMIC-IV	ICU Mort.	0.895±0.004, 0.533±0.018	0.737±0.070, 0.239±0.051	0.861±0.005, 0.407±0.034	0.746±0.058, 0.252±0.040	0.675±0.119, 0.235±0.087
MIMIC-IV	ICU Readmit	0.644±0.022, 0.299±0.020	0.490±0.019, 0.191±0.010	0.609±0.058, 0.265±0.048	0.510±0.010, 0.196±0.005	0.518±0.023, 0.201±0.014
eICU	COPD Mort.	0.794±0.010, 0.365±0.009	0.556±0.088, 0.157±0.060	0.749±0.015, 0.280±0.019	0.596±0.057, 0.153±0.044	0.547±0.069, 0.154±0.038
eICU	HF Mortality	0.739±0.016, 0.327±0.015	0.612±0.042, 0.194±0.032	0.714±0.011, 0.285±0.026	0.598±0.036, 0.186±0.023	0.661±0.062, 0.246±0.055
eICU	Length-of-Stay	0.755±0.008, 0.284±0.012	0.692±0.033, 0.218±0.024	0.723±0.011, 0.239±0.011	0.694±0.033, 0.223±0.021	0.648±0.057, 0.199±0.044
eICU	Mortality	0.898±0.006, 0.438±0.028	0.870±0.016, 0.380±0.021	0.849±0.028, 0.359±0.051	0.870±0.012, 0.375±0.020	0.737±0.114, 0.222±0.109
eICU	Readmit	0.695±0.022, 0.358±0.042	0.595±0.022, 0.205±0.019	0.609±0.013, 0.213±0.007	0.576±0.021, 0.197±0.017	0.520±0.050, 0.173±0.029
<i>Multiclass</i>						
SUPPORT2	Functional	0.822±0.010, 0.459±0.008	0.578±0.027, 0.274±0.018	0.771±0.013, 0.406±0.007	0.576±0.022, 0.303±0.007	0.600±0.029, 0.321±0.016
UCI Dermatology	Dermatological Dis.	0.996±0.003, 0.982±0.011	0.856±0.023, 0.593±0.039	0.997±0.002, 0.987±0.008	0.866±0.020, 0.576±0.039	0.740±0.040, 0.434±0.065
UCI Diabetes-130	Readmit (3-cl)	0.669±0.015, 0.468±0.015	0.570±0.011, 0.384±0.007	0.614±0.007, 0.418±0.003	0.568±0.008, 0.380±0.005	0.607±0.016, 0.409±0.007

Table 3: Per-dataset downstream utility. Each cell reports **AUROC±std**, **AUPRC±std** (mean ± std over the five classifiers; Real=TRTR, generators=TSTR). Best synthetic generator per row (by AUROC) in **bold**. Binary tasks use standard AUROC/AUPRC; the *Multiclass* block uses macro one-vs-rest AUROC and macro AUPRC.

cohorts.

Our central finding concerns usefulness rather than appearance. Synthetic tabular data preserves most downstream signal (mean AUC-ROC utility 90.6% for CoMed-CTGAN, up to 97.3% for CoMed-TVAE), and several datasets match the real-data upper bound. Temporal ICU tasks are harder and more generator-sensitive: under CoMed-CTGAN utility falls to 81.6% under AUC-ROC and to 64.0% under the imbalance-sensitive AUC-PRC, though CoMed-TVAE still retains ~95% under AUC-ROC. Relating the two axes, fidelity-especially the preservation of pairwise correlation structure-is positively but imperfectly correlated with utility (for CoMed-CTGAN, Overall $r = 0.67$, $\rho = 0.74$), while

dataset size is not. Statistical fidelity is therefore a useful but insufficient proxy, and downstream utility must be evaluated directly.

The practical message is that synthetic data is already a credible substitute for static tabular prototyping and benchmarking, while rare-event imbalance remains the hardest case. Because our pipeline summarizes each ICU stay into a per-stay feature vector, claims about fine-grained temporal dynamics are out of scope; this motivates comparison against sequence-native generators. Integrating privacy tests would complete the fidelity-utility-privacy picture needed for clinical deployment.

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CoMedBench: A Multi-Source Benchmark of Synthetic Medical Data Fidelity and Downstream Utility

Supplementary Material

This supplement provides the complete, non-aggregated benchmark results that the main paper summarizes for space. Results are reported at full granularity: one row per dataset×generator for statistical fidelity, and one row per dataset×task×classifier for downstream utility. Generators: CoMed-CTGAN, CoMed-TVAE, CoMed-CopulaGan(CoMed-CG), CoMed-GaussianCopula(CoMed-GC). Downstream classifiers: Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), XGBoost, MLP. Overall Quality is the mean of Distribution Stability and Correlation Stability. Utility: TRTR = train-real/test-real, TSTR = train-synthetic/test-real, and Retention (Ret.) = TSTR/TRTR. All numerical values match the experiments reported in the main paper; “–” denotes a run that is not available.

1 Datasets and Prediction Tasks

Table 1: Tabular datasets and prediction tasks.

Dataset	Task	Task type	Rows	#num / #cat
Breast Cancer Coimbra	Breast Cancer	Binary	116	9 / 0
CDC BRFSS	Diabetes	Binary	253,680	3 / 18
GBSG (pycox)	Breast Cancer Survival	Binary	2,232	4 / 3
METABRIC (pycox)	Breast Cancer Survival	Binary	1,904	5 / 4
SUPPORT2	Death	Binary	9,105	24 / 13
SUPPORT2	Functional Outcome	Multi-class (5)	7,705	24 / 13
SUPPORT2	Mortality	Binary	9,105	24 / 13
UCI Cervical Cancer	Cervical Cancer	Binary	858	6 / 22
UCI Chronic Kidney Disease	Chronic Kidney Disease	Binary	400	11 / 13
UCI Dermatology	Dermatological Disease Classification	Multi-class (6)	366	1 / 33
UCI Diabetes 130-US Hospitals	30-day Readmission	Binary	101,766	8 / 36
UCI Diabetes 130-US Hospitals	Any-Readmission	Binary	101,766	8 / 36
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Multi-class (3)	101,766	8 / 36
UCI Early-Stage Diabetes	Early Diabetes Risk	Binary	520	1 / 15
UCI Heart Failure	Heart Failure Mortality	Binary	299	6 / 5
UCI Hepatitis	Hepatitis Mortality	Binary	155	6 / 13
UCI Indian Liver Patient	Liver Disease	Binary	583	9 / 1
UCI Mammographic Mass	Mass Malignancy	Binary	961	1 / 4
UCI Statlog Heart	Heart Disease	Binary	270	5 / 8
NHANES	Mortality	Binary	33,654	70 / 81

Table 2: Time-series (ICU) datasets and prediction tasks.

Dataset	Task	Task type	Rows	#num / #cat
MIMIC-III	COPD Mortality	Binary	3,285	109 / 2
MIMIC-III	Heart Failure Mortality	Binary	1,176	38 / 10
MIMIC-III	Hospital Readmission	Binary	29,000	109 / 2
MIMIC-III	Length-of-Stay	Binary	32,551	109 / 2
MIMIC-III	ICU Readmission	Binary	29,000	109 / 2
MIMIC-III	Hospital Mortality	Binary	14,681	168 / 3
MIMIC-IV	COPD Mortality	Binary	6,385	126 / 3
MIMIC-IV	Heart Failure Mortality	Binary	13,838	130 / 3
MIMIC-IV	Hospital Readmission	Binary	59,646	34 / 249
MIMIC-IV	Length-of-Stay	Binary	74,829	39 / 4
MIMIC-IV	ICU Mortality	Binary	51,838	126 / 3
MIMIC-IV	ICU Readmission	Binary	9,648	62 / 280
eICU	COPD Mortality	Binary	7,735	35 / 9
eICU	Heart Failure Mortality	Binary	10,377	35 / 9
eICU	Length-of-Stay	Binary	91,852	34 / 9
eICU	Mortality	Binary	111,390	34 / 28
eICU	Readmission	Binary	104,639	104 / 9

2 Statistical Fidelity: Full Per-Dataset Results

Table 3: Per-dataset tabular statistical fidelity (SDV) for the four generators. Best Overall Quality per dataset in **bold**.

Dataset	Task	Generator	Distribution Stability	Correlation Stability	Overall Quality	Data Validity	Data Structure
Breast Cancer Coimbra	Breast Cancer	CoMed-CTGAN	0.828	0.661	0.744	1.000	1.000
Breast Cancer Coimbra	Breast Cancer	CoMed-TVAE	0.860	0.925	0.893	1.000	1.000
Breast Cancer Coimbra	Breast Cancer	CoMed-CG	0.812	0.708	0.760	1.000	1.000
Breast Cancer Coimbra	Breast Cancer	CoMed-GC	0.828	0.963	0.895	1.000	1.000
CDC BRFSS	Diabetes	CoMed-CTGAN	0.953	0.988	0.970	1.000	1.000
CDC BRFSS	Diabetes	CoMed-TVAE	0.953	0.928	0.940	1.000	1.000
CDC BRFSS	Diabetes	CoMed-CG	0.905	0.975	0.940	1.000	1.000
CDC BRFSS	Diabetes	CoMed-GC	0.934	0.765	0.849	1.000	1.000
GBSG (pycox)	Breast Cancer Survival	CoMed-CTGAN	0.931	0.939	0.935	1.000	1.000
GBSG (pycox)	Breast Cancer Survival	CoMed-TVAE	0.935	0.999	0.967	1.000	1.000
GBSG (pycox)	Breast Cancer Survival	CoMed-CG	0.909	0.909	0.909	1.000	1.000
GBSG (pycox)	Breast Cancer Survival	CoMed-GC	0.913	0.854	0.883	1.000	1.000
METABRIC (pycox)	Breast Cancer Survival	CoMed-CTGAN	0.935	0.970	0.953	1.000	1.000
METABRIC (pycox)	Breast Cancer Survival	CoMed-TVAE	0.968	0.990	0.979	1.000	1.000
METABRIC (pycox)	Breast Cancer Survival	CoMed-CG	0.900	0.958	0.929	1.000	1.000
METABRIC (pycox)	Breast Cancer Survival	CoMed-GC	0.956	0.859	0.907	1.000	1.000
SUPPORT2	Death	CoMed-CTGAN	0.840	0.816	0.828	0.998	1.000
SUPPORT2	Death	CoMed-TVAE	0.860	0.872	0.866	1.000	1.000
SUPPORT2	Death	CoMed-CG	0.835	0.814	0.825	1.000	1.000
SUPPORT2	Death	CoMed-GC	0.886	0.847	0.867	1.000	1.000
SUPPORT2	Functional Outcome	CoMed-CTGAN	0.843	0.789	0.816	1.000	1.000
SUPPORT2	Functional Outcome	CoMed-TVAE	0.861	0.898	0.879	1.000	1.000
SUPPORT2	Functional Outcome	CoMed-CG	0.873	0.822	0.847	1.000	1.000
SUPPORT2	Functional Outcome	CoMed-GC	0.889	0.861	0.875	1.000	1.000
SUPPORT2	Mortality	CoMed-CTGAN	0.846	0.842	0.846	1.000	1.000
SUPPORT2	Mortality	CoMed-TVAE	0.859	0.886	0.873	1.000	1.000
SUPPORT2	Mortality	CoMed-CG	0.825	0.796	0.810	1.000	1.000
SUPPORT2	Mortality	CoMed-GC	0.887	0.839	0.863	1.000	1.000
UCI Cervical Cancer	Cervical Cancer	CoMed-CTGAN	0.899	0.834	0.867	1.000	1.000
UCI Cervical Cancer	Cervical Cancer	CoMed-TVAE	0.946	0.881	0.913	1.000	1.000
UCI Cervical Cancer	Cervical Cancer	CoMed-CG	0.853	0.803	0.828	1.000	1.000
UCI Cervical Cancer	Cervical Cancer	CoMed-GC	0.885	0.869	0.877	1.000	1.000
UCI Chronic Kidney Disease	Chronic Kidney Disease	CoMed-CTGAN	0.883	0.808	0.846	1.000	1.000
UCI Chronic Kidney Disease	Chronic Kidney Disease	CoMed-TVAE	0.919	0.821	0.870	1.000	1.000
UCI Chronic Kidney Disease	Chronic Kidney Disease	CoMed-CG	0.904	0.771	0.838	1.000	1.000
UCI Chronic Kidney Disease	Chronic Kidney Disease	CoMed-GC	0.901	0.736	0.819	1.000	1.000
UCI Dermatology	Dermatological Disease Classification	CoMed-CTGAN	0.934	0.877	0.905	1.000	1.000
UCI Dermatology	Dermatological Disease Classification	CoMed-TVAE	0.971	0.961	0.966	1.000	1.000
UCI Dermatology	Dermatological Disease Classification	CoMed-CG	0.917	0.881	0.899	1.000	1.000
UCI Dermatology	Dermatological Disease Classification	CoMed-GC	0.973	0.709	0.841	1.000	1.000
UCI Diabetes 130-US Hospitals	30-day Readmission	CoMed-CTGAN	0.936	0.653	0.794	0.997	1.000
UCI Diabetes 130-US Hospitals	30-day Readmission	CoMed-TVAE	0.892	0.595	0.743	0.999	1.000
UCI Diabetes 130-US Hospitals	30-day Readmission	CoMed-CG	0.897	0.680	0.788	0.997	1.000
UCI Diabetes 130-US Hospitals	30-day Readmission	CoMed-GC	0.921	0.729	0.825	0.998	1.000

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Table 3 (continued)

Dataset	Task	Generator	Distribution Stability	Correlation Stability	Overall Quality	Data Validity	Data Structure
UCI Diabetes 130-US Hospitals	Any-Readmission	CoMed-CTGAN	0.938	0.652	0.795	0.997	1.000
UCI Diabetes 130-US Hospitals	Any-Readmission	CoMed-TVAE	0.898	0.632	0.765	0.999	1.000
UCI Diabetes 130-US Hospitals	Any-Readmission	CoMed-CG	0.897	0.675	0.786	0.997	1.000
UCI Diabetes 130-US Hospitals	Any-Readmission	CoMed-GC	0.921	0.734	0.827	0.998	1.000
UCI Diabetes 130-US Hospitals	Readmission (3-class)	CoMed-CTGAN	0.930	0.640	0.785	0.997	1.000
UCI Diabetes 130-US Hospitals	Readmission (3-class)	CoMed-TVAE	0.897	0.632	0.765	0.999	1.000
UCI Diabetes 130-US Hospitals	Readmission (3-class)	CoMed-CG	0.890	0.636	0.763	0.996	1.000
UCI Diabetes 130-US Hospitals	Readmission (3-class)	CoMed-GC	0.921	0.728	0.824	0.998	1.000
UCI Early-Stage Diabetes	Early Diabetes Risk	CoMed-CTGAN	0.921	0.851	0.886	1.000	1.000
UCI Early-Stage Diabetes	Early Diabetes Risk	CoMed-TVAE	0.976	0.940	0.958	1.000	1.000
UCI Early-Stage Diabetes	Early Diabetes Risk	CoMed-CG	0.896	0.829	0.863	1.000	1.000
UCI Early-Stage Diabetes	Early Diabetes Risk	CoMed-GC	0.970	0.875	0.922	1.000	1.000
UCI Heart Failure	Heart Failure Mortality	CoMed-CTGAN	0.880	–	0.880	1.000	1.000
UCI Heart Failure	Heart Failure Mortality	CoMed-TVAE	0.929	–	0.929	1.000	1.000
UCI Heart Failure	Heart Failure Mortality	CoMed-CG	0.835	–	0.835	1.000	1.000
UCI Heart Failure	Heart Failure Mortality	CoMed-GC	0.922	–	0.922	1.000	1.000
UCI Hepatitis	Hepatitis Mortality	CoMed-CTGAN	0.873	0.919	0.896	1.000	1.000
UCI Hepatitis	Hepatitis Mortality	CoMed-TVAE	0.926	0.955	0.940	1.000	1.000
UCI Hepatitis	Hepatitis Mortality	CoMed-CG	0.867	0.901	0.884	1.000	1.000
UCI Hepatitis	Hepatitis Mortality	CoMed-GC	0.864	0.795	0.830	1.000	1.000
UCI Indian Liver Patient	Liver Disease	CoMed-CTGAN	0.854	0.805	0.830	1.000	1.000
UCI Indian Liver Patient	Liver Disease	CoMed-TVAE	0.914	0.958	0.936	1.000	1.000
UCI Indian Liver Patient	Liver Disease	CoMed-CG	0.890	0.819	0.855	1.000	1.000
UCI Indian Liver Patient	Liver Disease	CoMed-GC	0.873	0.962	0.918	1.000	1.000
UCI Mammographic Mass	Mass Malignancy	CoMed-CTGAN	0.928	0.974	0.951	1.000	1.000
UCI Mammographic Mass	Mass Malignancy	CoMed-TVAE	0.975	0.978	0.977	1.000	1.000
UCI Mammographic Mass	Mass Malignancy	CoMed-CG	0.953	0.927	0.940	1.000	1.000
UCI Mammographic Mass	Mass Malignancy	CoMed-GC	0.985	0.694	0.839	1.000	1.000
UCI Statlog Heart	Heart Disease	CoMed-CTGAN	0.887	0.939	0.903	1.000	1.000
UCI Statlog Heart	Heart Disease	CoMed-TVAE	0.940	0.952	0.946	1.000	1.000
UCI Statlog Heart	Heart Disease	CoMed-CG	0.866	0.861	0.863	1.000	1.000
UCI Statlog Heart	Heart Disease	CoMed-GC	0.879	0.780	0.829	1.000	1.000
NHANES	Mortality	CoMed-CTGAN	0.893	0.640	0.767	0.992	1.000
NHANES	Mortality	CoMed-TVAE	0.921	0.966	0.944	1.000	1.000
NHANES	Mortality	CoMed-CG	0.910	0.698	0.804	0.998	1.000
NHANES	Mortality	CoMed-GC	0.934	0.954	0.944	0.996	1.000

Table 4: Per-dataset time-series statistical fidelity (SDV) for the four generators. Best Overall Quality per dataset in **bold**.

Dataset	Task	Generator	Distribution Stability	Correlation Stability	Overall Quality	Data Validity	Data Structure
MIMIC-III	COPD Mortality	CoMed-CTGAN	0.860	0.733	0.796	1.000	1.000
MIMIC-III	COPD Mortality	CoMed-TVAE	0.876	0.910	0.893	1.000	1.000
MIMIC-III	COPD Mortality	CoMed-CG	0.849	0.732	0.791	1.000	1.000
MIMIC-III	COPD Mortality	CoMed-GC	0.846	0.963	0.904	1.000	1.000
MIMIC-III	Heart Failure Mortality	CoMed-CTGAN	0.821	0.688	0.754	1.000	1.000
MIMIC-III	Heart Failure Mortality	CoMed-TVAE	0.917	0.912	0.915	1.000	1.000
MIMIC-III	Heart Failure Mortality	CoMed-CG	0.807	0.691	0.749	1.000	1.000
MIMIC-III	Heart Failure Mortality	CoMed-GC	0.923	0.986	0.955	1.000	1.000
MIMIC-III	Hospital Readmission	CoMed-CTGAN	0.859	0.800	0.829	0.997	1.000
MIMIC-III	Hospital Readmission	CoMed-TVAE	0.836	0.948	0.892	1.000	1.000
MIMIC-III	Hospital Readmission	CoMed-CG	0.869	0.790	0.829	0.998	1.000
MIMIC-III	Hospital Readmission	CoMed-GC	0.828	0.921	0.875	0.993	1.000
MIMIC-III	Length-of-Stay	CoMed-CTGAN	0.867	0.812	0.840	0.998	1.000
MIMIC-III	Length-of-Stay	CoMed-TVAE	0.850	0.941	0.895	1.000	1.000
MIMIC-III	Length-of-Stay	CoMed-CG	0.870	0.811	0.841	0.998	1.000
MIMIC-III	Length-of-Stay	CoMed-GC	0.823	0.920	0.871	0.992	1.000
MIMIC-III	ICU Readmission	CoMed-CTGAN	0.871	0.800	0.836	0.997	1.000
MIMIC-III	ICU Readmission	CoMed-TVAE	0.842	0.941	0.892	1.000	1.000
MIMIC-III	ICU Readmission	CoMed-CG	0.866	0.797	0.832	0.997	1.000
MIMIC-III	ICU Readmission	CoMed-GC	0.828	0.921	0.875	0.993	1.000
MIMIC-III	Hospital Mortality	CoMed-CTGAN	0.835	0.857	0.846	0.993	1.000
MIMIC-III	Hospital Mortality	CoMed-TVAE	0.824	0.923	0.873	0.996	1.000
MIMIC-III	Hospital Mortality	CoMed-CG	0.842	0.853	0.847	0.995	1.000
MIMIC-III	Hospital Mortality	CoMed-GC	0.761	0.870	0.815	0.995	1.000
MIMIC-IV	COPD Mortality	CoMed-CTGAN	0.859	0.750	0.805	1.000	1.000
MIMIC-IV	COPD Mortality	CoMed-TVAE	0.845	0.925	0.885	1.000	1.000
MIMIC-IV	COPD Mortality	CoMed-CG	0.866	0.753	0.810	1.000	1.000
MIMIC-IV	COPD Mortality	CoMed-GC	0.798	0.918	0.858	1.000	1.000
MIMIC-IV	Heart Failure Mortality	CoMed-CTGAN	0.851	0.798	0.825	0.998	1.000
MIMIC-IV	Heart Failure Mortality	CoMed-TVAE	0.836	0.930	0.883	0.998	1.000
MIMIC-IV	Heart Failure Mortality	CoMed-CG	0.853	0.790	0.822	0.998	1.000
MIMIC-IV	Heart Failure Mortality	CoMed-GC	0.811	0.920	0.866	0.997	1.000
MIMIC-IV	Hospital Readmission	CoMed-CTGAN	0.895	0.795	0.845	0.997	1.000
MIMIC-IV	Hospital Readmission	CoMed-TVAE	0.930	0.927	0.928	1.000	1.000
MIMIC-IV	Hospital Readmission	CoMed-CG	0.931	0.861	0.896	0.999	1.000
MIMIC-IV	Hospital Readmission	CoMed-GC	0.969	0.862	0.915	1.000	1.000
MIMIC-IV	Length-of-Stay	CoMed-CTGAN	0.875	0.957	0.916	0.999	1.000
MIMIC-IV	Length-of-Stay	CoMed-TVAE	0.870	0.934	0.902	0.994	1.000
MIMIC-IV	Length-of-Stay	CoMed-CG	0.878	0.948	0.913	0.997	1.000
MIMIC-IV	Length-of-Stay	CoMed-GC	0.884	0.969	0.926	0.980	1.000
MIMIC-IV	ICU Mortality	CoMed-CTGAN	0.857	0.855	0.856	0.999	1.000
MIMIC-IV	ICU Mortality	CoMed-TVAE	0.835	0.946	0.890	1.000	1.000
MIMIC-IV	ICU Mortality	CoMed-CG	0.862	0.853	0.857	0.999	1.000
MIMIC-IV	ICU Mortality	CoMed-GC	0.796	0.893	0.845	0.995	1.000

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Table 4 (continued)

Dataset	Task	Generator	Distribution Stability	Correlation Stability	Overall Quality	Data Validity	Data Structure
MIMIC-IV	ICU Readmission	CoMed-CTGAN	0.844	0.664	0.754	0.999	1.000
MIMIC-IV	ICU Readmission	CoMed-TVAE	0.905	0.883	0.894	1.000	1.000
MIMIC-IV	ICU Readmission	CoMed-CG	0.830	0.668	0.749	0.999	1.000
MIMIC-IV	ICU Readmission	CoMed-GC	0.950	0.874	0.912	1.000	1.000
eICU	COPD Mortality	CoMed-CTGAN	0.815	0.786	0.800	1.000	1.000
eICU	COPD Mortality	CoMed-TVAE	0.800	0.880	0.840	1.000	1.000
eICU	COPD Mortality	CoMed-CG	0.809	0.808	0.809	1.000	1.000
eICU	COPD Mortality	CoMed-GC	0.829	0.902	0.865	1.000	1.000
eICU	Heart Failure Mortality	CoMed-CTGAN	0.815	0.848	0.832	1.000	1.000
eICU	Heart Failure Mortality	CoMed-TVAE	0.808	0.911	0.860	1.000	1.000
eICU	Heart Failure Mortality	CoMed-CG	0.795	0.843	0.819	1.000	1.000
eICU	Heart Failure Mortality	CoMed-GC	0.835	0.909	0.872	1.000	1.000
eICU	Length-of-Stay	CoMed-CTGAN	0.796	0.919	0.857	1.000	1.000
eICU	Length-of-Stay	CoMed-TVAE	0.791	0.935	0.863	1.000	1.000
eICU	Length-of-Stay	CoMed-CG	0.808	0.921	0.865	1.000	1.000
eICU	Length-of-Stay	CoMed-GC	0.842	0.883	0.863	0.998	1.000
eICU	Mortality	CoMed-CTGAN	0.871	0.943	0.907	1.000	1.000
eICU	Mortality	CoMed-TVAE	0.853	0.914	0.884	1.000	1.000
eICU	Mortality	CoMed-CG	0.885	0.944	0.915	1.000	1.000
eICU	Mortality	CoMed-GC	0.901	0.820	0.860	0.993	1.000
eICU	Readmission	CoMed-CTGAN	0.823	0.903	0.863	1.000	1.000
eICU	Readmission	CoMed-TVAE	0.814	0.967	0.890	1.000	1.000
eICU	Readmission	CoMed-CG	0.812	0.895	0.854	1.000	1.000
eICU	Readmission	CoMed-GC	0.818	0.951	0.885	0.998	1.000

3 Downstream Utility: Tabular (Per-Classifer)

Table 5: Per-dataset, per-classifier tabular downstream utility, AUROC (with the clinical-validity layer). Real = TRTR; each generator reports TSTR and Retention = TSTR/TRTR.

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
Breast Cancer Coimbra	Breast Cancer	Logistic Regression	–	0.776	0.762 (98.2%)	0.678 (87.4%)	0.769 (99.1%)	0.615 (79.3%)
Breast Cancer Coimbra	Breast Cancer	Random Forest	–	0.846	0.647 (76.5%)	0.783 (92.6%)	0.573 (67.8%)	0.787 (93.0%)
Breast Cancer Coimbra	Breast Cancer	Gradient Boosting	–	0.881	0.622 (70.6%)	0.776 (88.1%)	0.587 (66.7%)	0.755 (85.7%)
Breast Cancer Coimbra	Breast Cancer	XGBoost	–	0.818	0.678 (82.9%)	0.776 (94.9%)	0.608 (74.4%)	0.818 (100.0%)
Breast Cancer Coimbra	Breast Cancer	MLP	–	0.566	0.378 (66.7%)	0.643 (113.6%)	0.692 (122.2%)	0.573 (101.2%)
CDC BRFS	Diabetes	Logistic Regression	–	0.820	0.812 (99.0%)	0.815 (99.5%)	0.768 (93.7%)	0.799 (97.5%)
CDC BRFS	Diabetes	Random Forest	–	0.798	0.775 (97.1%)	0.791 (99.1%)	0.739 (92.6%)	0.715 (89.6%)
CDC BRFS	Diabetes	Gradient Boosting	–	0.825	0.809 (98.0%)	0.817 (99.0%)	0.779 (94.4%)	0.785 (95.1%)
CDC BRFS	Diabetes	XGBoost	–	0.824	0.799 (96.9%)	0.809 (98.1%)	0.732 (88.7%)	0.722 (87.6%)
CDC BRFS	Diabetes	MLP	–	0.826	0.806 (97.6%)	0.814 (98.6%)	0.755 (91.4%)	0.770 (93.3%)

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Table 5 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
GBSG (pycox)	Breast Cancer Survival	Logistic Regression	–	0.682	0.677 (99.3%)	0.673 (98.7%)	0.667 (97.8%)	0.669 (98.2%)
GBSG (pycox)	Breast Cancer Survival	Random Forest	–	0.644	0.657 (102.0%)	0.667 (103.5%)	0.652 (101.2%)	0.660 (102.4%)
GBSG (pycox)	Breast Cancer Survival	Gradient Boosting	–	0.681	0.682 (100.1%)	0.683 (100.2%)	0.651 (95.6%)	0.653 (95.8%)
GBSG (pycox)	Breast Cancer Survival	XGBoost	–	0.634	0.632 (99.7%)	0.682 (107.5%)	0.631 (99.5%)	0.667 (105.1%)
GBSG (pycox)	Breast Cancer Survival	MLP	–	0.674	0.684 (101.5%)	0.676 (100.2%)	0.668 (99.2%)	0.657 (97.5%)
METABRIC (pycox)	Breast Cancer Survival	Logistic Regression	–	0.737	0.689 (93.5%)	0.739 (100.4%)	0.692 (94.0%)	0.742 (100.8%)
METABRIC (pycox)	Breast Cancer Survival	Random Forest	–	0.722	0.647 (89.7%)	0.694 (96.2%)	0.683 (94.6%)	0.686 (95.1%)
METABRIC (pycox)	Breast Cancer Survival	Gradient Boosting	–	0.720	0.662 (92.0%)	0.705 (98.0%)	0.704 (97.9%)	0.706 (98.2%)
METABRIC (pycox)	Breast Cancer Survival	XGBoost	–	0.699	0.637 (91.1%)	0.661 (94.6%)	0.675 (96.6%)	0.629 (90.0%)
METABRIC (pycox)	Breast Cancer Survival	MLP	–	0.726	0.700 (96.3%)	0.743 (102.2%)	0.694 (95.6%)	0.716 (98.5%)
SUPPORT2	Death	Logistic Regression	–	0.841	0.711 (84.6%)	0.819 (97.5%)	0.707 (84.1%)	0.772 (91.9%)
SUPPORT2	Death	Random Forest	–	0.849	0.636 (74.9%)	0.830 (97.7%)	0.653 (76.9%)	0.763 (89.9%)
SUPPORT2	Death	Gradient Boosting	–	0.855	0.632 (73.8%)	0.829 (96.9%)	0.626 (73.2%)	0.764 (89.3%)
SUPPORT2	Death	XGBoost	–	0.847	0.613 (72.4%)	0.827 (97.6%)	0.638 (75.3%)	0.751 (88.7%)
SUPPORT2	Death	MLP	–	0.846	0.697 (82.3%)	0.830 (98.1%)	0.675 (79.8%)	0.769 (90.9%)
SUPPORT2	Functional Outcome	Logistic Regression	–	0.807	0.555 (68.7%)	0.748 (92.7%)	0.555 (68.7%)	0.578 (71.6%)
SUPPORT2	Functional Outcome	Random Forest	–	0.821	0.607 (73.9%)	0.773 (94.2%)	0.598 (72.8%)	0.634 (77.2%)
SUPPORT2	Functional Outcome	Gradient Boosting	–	0.833	0.602 (72.3%)	0.780 (93.5%)	0.600 (72.0%)	0.623 (74.8%)
SUPPORT2	Functional Outcome	XGBoost	–	0.823	0.580 (70.5%)	0.774 (94.0%)	0.574 (69.7%)	0.596 (72.4%)
SUPPORT2	Functional Outcome	MLP	–	0.826	0.545 (66.0%)	0.782 (94.6%)	0.554 (67.0%)	0.567 (68.6%)
SUPPORT2	Mortality	Logistic Regression	–	0.946	0.977 (92.7%)	0.936 (98.9%)	0.869 (91.8%)	0.900 (95.1%)
SUPPORT2	Mortality	Random Forest	–	0.949	0.861 (90.7%)	0.936 (98.6%)	0.823 (86.7%)	0.850 (89.6%)
SUPPORT2	Mortality	Gradient Boosting	–	0.951	0.848 (89.2%)	0.934 (98.2%)	0.827 (86.9%)	0.852 (89.6%)
SUPPORT2	Mortality	XGBoost	–	0.950	0.785 (82.6%)	0.925 (97.4%)	0.766 (80.7%)	0.677 (71.3%)
SUPPORT2	Mortality	MLP	–	0.951	0.863 (90.8%)	0.932 (98.0%)	0.834 (87.7%)	0.842 (88.6%)
UCI Cervical Cancer	Cervical Cancer	Logistic Regression	–	0.574	0.533 (92.9%)	0.406 (70.8%)	0.452 (78.7%)	0.522 (91.0%)
UCI Cervical Cancer	Cervical Cancer	Random Forest	–	0.698	0.586 (83.9%)	0.461 (66.0%)	0.476 (68.2%)	0.529 (75.8%)
UCI Cervical Cancer	Cervical Cancer	Gradient Boosting	–	0.665	0.577 (86.7%)	0.522 (78.5%)	0.626 (94.1%)	0.556 (83.6%)
UCI Cervical Cancer	Cervical Cancer	XGBoost	–	0.651	0.513 (78.8%)	0.481 (73.9%)	0.510 (78.4%)	0.465 (71.5%)
UCI Cervical Cancer	Cervical Cancer	MLP	–	0.427	0.456 (106.9%)	0.490 (114.7%)	0.402 (94.2%)	0.410 (96.0%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	Logistic Regression	–	1.000	0.987 (98.7%)	1.000 (100.0%)	0.989 (98.9%)	0.981 (98.1%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	Random Forest	–	1.000	0.993 (99.3%)	1.000 (100.0%)	0.997 (99.7%)	0.981 (98.1%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	Gradient Boosting	–	1.000	0.992 (99.2%)	0.999 (99.9%)	0.992 (99.2%)	0.922 (92.2%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	XGBoost	–	0.999	0.983 (98.5%)	1.000 (100.1%)	0.989 (99.1%)	0.919 (92.1%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	MLP	–	0.995	0.991 (99.6%)	0.998 (100.3%)	0.984 (98.9%)	0.996 (100.1%)
UCI Dermatology	Dermatological Disease Classification	Logistic Regression	–	0.999	0.834 (83.4%)	0.995 (99.5%)	0.853 (85.3%)	0.781 (78.2%)
UCI Dermatology	Dermatological Disease Classification	Random Forest	–	0.998	0.872 (87.4%)	1.000 (100.1%)	0.856 (85.7%)	0.779 (78.1%)
UCI Dermatology	Dermatological Disease Classification	Gradient Boosting	–	0.995	0.872 (87.7%)	0.997 (100.2%)	0.871 (87.6%)	0.709 (71.2%)
UCI Dermatology	Dermatological Disease Classification	XGBoost	–	0.995	0.874 (87.8%)	0.998 (100.3%)	0.852 (85.6%)	0.736 (73.9%)
UCI Dermatology	Dermatological Disease Classification	MLP	–	0.994	0.830 (83.5%)	0.996 (100.2%)	0.898 (90.4%)	0.693 (69.8%)
UCI Diabetes 130-US Hospitals	30-day Readmission	Logistic Regression	–	0.653	0.519 (79.5%)	0.571 (87.3%)	0.558 (85.4%)	0.630 (96.5%)
UCI Diabetes 130-US Hospitals	30-day Readmission	Random Forest	–	0.670	0.513 (76.6%)	0.553 (82.5%)	0.546 (81.4%)	0.596 (88.9%)
UCI Diabetes 130-US Hospitals	30-day Readmission	Gradient Boosting	–	0.676	0.526 (77.8%)	0.538 (79.7%)	0.578 (85.5%)	0.582 (86.0%)
UCI Diabetes 130-US Hospitals	30-day Readmission	XGBoost	–	0.672	0.506 (75.4%)	0.571 (85.0%)	0.516 (76.8%)	0.553 (82.4%)
UCI Diabetes 130-US Hospitals	30-day Readmission	MLP	–	0.661	0.503 (76.1%)	0.609 (92.2%)	0.541 (81.9%)	0.604 (91.3%)

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Table 5 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
UCI Diabetes 130-US Hospitals	Any-Readmission	Logistic Regression	–	0.669	0.619 (92.6%)	0.652 (97.5%)	0.620 (92.6%)	0.654 (97.8%)
UCI Diabetes 130-US Hospitals	Any-Readmission	Random Forest	–	0.697	0.606 (87.0%)	0.653 (93.7%)	0.598 (85.8%)	0.636 (91.3%)
UCI Diabetes 130-US Hospitals	Any-Readmission	Gradient Boosting	–	0.692	0.625 (90.2%)	0.648 (93.7%)	0.619 (89.5%)	0.636 (91.9%)
UCI Diabetes 130-US Hospitals	Any-Readmission	XGBoost	–	0.706	0.607 (86.0%)	0.655 (92.7%)	0.595 (84.3%)	0.633 (89.6%)
UCI Diabetes 130-US Hospitals	Any-Readmission	MLP	–	0.688	0.614 (89.3%)	0.656 (95.4%)	0.622 (90.4%)	0.650 (94.6%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Logistic Regression	–	0.646	0.561 (86.8%)	0.610 (94.3%)	0.570 (88.3%)	0.579 (89.7%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Random Forest	–	0.672	0.566 (84.3%)	0.617 (91.8%)	0.569 (84.7%)	0.611 (91.0%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Gradient Boosting	–	0.675	0.589 (87.2%)	0.614 (90.9%)	0.555 (82.2%)	0.621 (92.0%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	XGBoost	–	0.687	0.568 (82.7%)	0.625 (91.0%)	0.571 (83.1%)	0.611 (88.9%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	MLP	–	0.667	0.566 (84.8%)	0.607 (90.9%)	0.577 (86.5%)	0.615 (92.2%)
UCI Early-Stage Diabetes	Early Diabetes Risk	Logistic Regression	–	0.990	0.990 (99.9%)	0.993 (100.3%)	0.971 (98.0%)	0.963 (97.2%)
UCI Early-Stage Diabetes	Early Diabetes Risk	Random Forest	–	0.998	0.988 (99.0%)	0.993 (99.5%)	0.961 (96.2%)	0.897 (89.9%)
UCI Early-Stage Diabetes	Early Diabetes Risk	Gradient Boosting	–	0.994	0.991 (99.8%)	0.984 (99.1%)	0.965 (97.1%)	0.918 (92.4%)
UCI Early-Stage Diabetes	Early Diabetes Risk	XGBoost	–	1.000	0.987 (98.7%)	0.983 (98.3%)	0.960 (96.0%)	0.893 (89.3%)
UCI Early-Stage Diabetes	Early Diabetes Risk	MLP	–	0.988	0.985 (99.6%)	0.932 (94.3%)	0.956 (96.7%)	0.913 (92.4%)
UCI Heart Failure	Heart Failure Mortality	Logistic Regression	0.643	0.734	0.673 (91.6%)	0.750 (102.1%)	0.783 (106.6%)	0.738 (100.5%)
UCI Heart Failure	Heart Failure Mortality	Random Forest	0.800	0.791	0.648 (81.9%)	0.808 (102.2%)	0.801 (101.3%)	0.709 (89.6%)
UCI Heart Failure	Heart Failure Mortality	Gradient Boosting	0.754	0.747	0.706 (94.5%)	0.795 (106.4%)	0.777 (104.0%)	0.573 (76.6%)
UCI Heart Failure	Heart Failure Mortality	XGBoost	0.754	0.769	0.661 (86.0%)	0.820 (106.7%)	0.736 (95.7%)	0.662 (86.1%)
UCI Heart Failure	Heart Failure Mortality	MLP	0.559	0.827	0.766 (92.7%)	0.815 (98.6%)	0.743 (89.9%)	0.770 (93.2%)
UCI Hepatitis	Hepatitis Mortality	Logistic Regression	–	0.867	0.847 (97.7%)	0.840 (96.9%)	0.443 (51.1%)	0.273 (31.5%)
UCI Hepatitis	Hepatitis Mortality	Random Forest	–	0.903	0.893 (98.9%)	0.920 (101.8%)	0.853 (94.5%)	0.807 (89.3%)
UCI Hepatitis	Hepatitis Mortality	Gradient Boosting	–	0.907	0.860 (94.8%)	0.853 (94.1%)	0.747 (82.3%)	0.853 (94.1%)
UCI Hepatitis	Hepatitis Mortality	XGBoost	–	0.887	0.933 (105.3%)	0.900 (101.5%)	0.700 (79.0%)	0.800 (90.2%)
UCI Hepatitis	Hepatitis Mortality	MLP	–	0.827	0.800 (96.8%)	0.833 (100.8%)	0.767 (92.7%)	0.533 (64.5%)
UCI Indian Liver Patient	Liver Disease	Logistic Regression	–	0.733	0.670 (91.3%)	0.745 (101.6%)	0.610 (83.1%)	0.779 (106.2%)
UCI Indian Liver Patient	Liver Disease	Random Forest	–	0.676	0.683 (101.1%)	0.744 (110.1%)	0.564 (83.5%)	0.619 (91.7%)
UCI Indian Liver Patient	Liver Disease	Gradient Boosting	–	0.648	0.635 (98.0%)	0.720 (111.0%)	0.610 (94.0%)	0.658 (101.5%)
UCI Indian Liver Patient	Liver Disease	XGBoost	–	0.636	0.649 (102.1%)	0.727 (114.2%)	0.582 (91.4%)	0.560 (88.0%)
UCI Indian Liver Patient	Liver Disease	MLP	–	0.733	0.706 (96.3%)	0.733 (100.0%)	0.648 (88.3%)	0.752 (102.6%)
UCI Mammographic Mass	Mass Malignancy	Logistic Regression	–	0.902	0.878 (97.4%)	0.908 (100.7%)	0.876 (97.1%)	0.786 (87.2%)
UCI Mammographic Mass	Mass Malignancy	Random Forest	–	0.863	0.867 (100.4%)	0.889 (103.0%)	0.847 (98.1%)	0.601 (69.6%)
UCI Mammographic Mass	Mass Malignancy	Gradient Boosting	–	0.894	0.915 (102.4%)	0.910 (101.8%)	0.891 (99.7%)	0.740 (82.8%)
UCI Mammographic Mass	Mass Malignancy	XGBoost	–	0.875	0.883 (101.0%)	0.899 (102.8%)	0.873 (99.8%)	0.593 (67.8%)
UCI Mammographic Mass	Mass Malignancy	MLP	–	0.894	0.887 (99.1%)	0.904 (101.0%)	0.880 (98.4%)	0.781 (87.3%)
UCI Statlog Heart	Heart Disease	Logistic Regression	–	0.896	0.858 (95.8%)	0.932 (104.0%)	0.874 (97.5%)	0.569 (63.6%)
UCI Statlog Heart	Heart Disease	Random Forest	–	0.871	0.871 (100.0%)	0.901 (103.5%)	0.850 (97.6%)	0.828 (95.1%)
UCI Statlog Heart	Heart Disease	Gradient Boosting	–	0.885	0.810 (91.5%)	0.904 (102.2%)	0.812 (91.8%)	0.703 (79.4%)
UCI Statlog Heart	Heart Disease	XGBoost	–	0.857	0.831 (96.9%)	0.908 (106.0%)	0.846 (98.7%)	0.807 (94.2%)
UCI Statlog Heart	Heart Disease	MLP	–	0.893	0.889 (99.5%)	0.881 (98.6%)	0.849 (95.0%)	0.439 (49.1%)
NHANES	Mortality	Logistic Regression	–	0.923	0.593 (64.3%)	0.892 (96.6%)	0.804 (87.1%)	0.907 (98.2%)
NHANES	Mortality	Random Forest	–	0.914	0.508 (55.5%)	0.907 (99.1%)	0.754 (82.4%)	0.894 (97.7%)
NHANES	Mortality	Gradient Boosting	–	0.923	0.596 (64.5%)	0.908 (98.4%)	0.772 (83.6%)	0.905 (98.0%)
NHANES	Mortality	XGBoost	–	0.923	0.466 (50.5%)	0.915 (99.2%)	0.732 (79.4%)	0.835 (90.5%)
NHANES	Mortality	MLP	–	0.915	0.555 (60.6%)	0.844 (92.2%)	0.783 (85.5%)	0.798 (87.2%)

Table 6: Per-dataset, per-classifier tabular downstream utility, AUPRC (with the clinical-validity layer).

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
Breast Cancer Coimbra	Breast Cancer	Logistic Regression	–	0.831	0.763 (91.9%)	0.743 (89.4%)	0.827 (99.5%)	0.705 (84.8%)
Breast Cancer Coimbra	Breast Cancer	Random Forest	–	0.811	0.664 (81.9%)	0.792 (97.6%)	0.725 (89.4%)	0.844 (104.0%)
Breast Cancer Coimbra	Breast Cancer	Gradient Boosting	–	0.807	0.667 (82.6%)	0.827 (102.4%)	0.750 (92.9%)	0.839 (104.0%)
Breast Cancer Coimbra	Breast Cancer	XGBoost	–	0.754	0.740 (98.2%)	0.803 (106.6%)	0.766 (101.6%)	0.881 (116.9%)
Breast Cancer Coimbra	Breast Cancer	MLP	–	0.651	0.523 (80.2%)	0.724 (111.0%)	0.690 (105.9%)	0.636 (97.6%)
CDC BRFSS	Diabetes	Logistic Regression	–	0.393	0.381 (97.1%)	0.388 (98.7%)	0.326 (83.0%)	0.367 (93.4%)
CDC BRFSS	Diabetes	Random Forest	–	0.351	0.337 (95.8%)	0.345 (98.3%)	0.274 (78.0%)	0.253 (71.9%)
CDC BRFSS	Diabetes	Gradient Boosting	–	0.420	0.392 (93.3%)	0.403 (96.0%)	0.336 (80.1%)	0.341 (81.2%)
CDC BRFSS	Diabetes	XGBoost	–	0.419	0.385 (91.9%)	0.380 (90.8%)	0.304 (72.7%)	0.304 (72.7%)
CDC BRFSS	Diabetes	MLP	–	0.420	0.392 (93.3%)	0.392 (93.5%)	0.305 (72.6%)	0.350 (83.4%)
GBSG (pycox)	Breast Cancer Survival	Logistic Regression	–	0.712	0.715 (100.4%)	0.706 (99.1%)	0.702 (98.5%)	0.702 (98.6%)
GBSG (pycox)	Breast Cancer Survival	Random Forest	–	0.670	0.703 (105.0%)	0.715 (106.8%)	0.698 (104.2%)	0.710 (106.0%)
GBSG (pycox)	Breast Cancer Survival	Gradient Boosting	–	0.705	0.717 (101.7%)	0.715 (101.5%)	0.691 (98.1%)	0.714 (101.3%)
GBSG (pycox)	Breast Cancer Survival	XGBoost	–	0.656	0.695 (105.9%)	0.728 (110.9%)	0.696 (106.0%)	0.716 (109.1%)
GBSG (pycox)	Breast Cancer Survival	MLP	–	0.711	0.720 (101.3%)	0.706 (99.3%)	0.706 (99.4%)	0.691 (97.2%)
METABRIC (pycox)	Breast Cancer Survival	Logistic Regression	–	0.815	0.762 (93.6%)	0.814 (100.0%)	0.763 (93.6%)	0.812 (99.7%)
METABRIC (pycox)	Breast Cancer Survival	Random Forest	–	0.777	0.701 (90.2%)	0.766 (98.5%)	0.745 (95.8%)	0.758 (97.5%)
METABRIC (pycox)	Breast Cancer Survival	Gradient Boosting	–	0.786	0.714 (90.8%)	0.781 (99.5%)	0.761 (96.9%)	0.763 (97.1%)
METABRIC (pycox)	Breast Cancer Survival	XGBoost	–	0.766	0.680 (88.7%)	0.744 (97.1%)	0.747 (97.5%)	0.722 (94.3%)
METABRIC (pycox)	Breast Cancer Survival	MLP	–	0.785	0.765 (97.4%)	0.801 (102.0%)	0.777 (98.9%)	0.765 (97.4%)
SUPPORT2	Death	Logistic Regression	–	0.918	0.823 (89.7%)	0.902 (98.3%)	0.838 (91.3%)	0.872 (95.0%)
SUPPORT2	Death	Random Forest	–	0.921	0.771 (83.8%)	0.906 (98.4%)	0.793 (86.2%)	0.867 (94.1%)
SUPPORT2	Death	Gradient Boosting	–	0.927	0.785 (84.7%)	0.905 (97.7%)	0.779 (84.1%)	0.869 (93.8%)
SUPPORT2	Death	XGBoost	–	0.924	0.770 (83.3%)	0.908 (98.2%)	0.783 (84.8%)	0.857 (92.7%)
SUPPORT2	Death	MLP	–	0.919	0.828 (90.1%)	0.911 (99.1%)	0.815 (88.7%)	0.875 (95.1%)
SUPPORT2	Functional Outcome	Logistic Regression	–	0.449	0.287 (64.0%)	0.400 (89.2%)	0.298 (66.3%)	0.302 (67.3%)
SUPPORT2	Functional Outcome	Random Forest	–	0.454	0.275 (60.7%)	0.413 (91.2%)	0.310 (68.4%)	0.320 (70.5%)
SUPPORT2	Functional Outcome	Gradient Boosting	–	0.469	0.257 (54.7%)	0.410 (87.5%)	0.301 (64.1%)	0.312 (66.5%)
SUPPORT2	Functional Outcome	XGBoost	–	0.459	0.256 (55.8%)	0.399 (86.9%)	0.296 (64.6%)	0.324 (70.6%)
SUPPORT2	Functional Outcome	MLP	–	0.462	0.296 (64.1%)	0.410 (88.6%)	0.311 (67.3%)	0.345 (74.7%)
SUPPORT2	Mortality	Logistic Regression	–	0.891	0.723 (81.1%)	0.861 (96.7%)	0.723 (81.1%)	0.782 (87.8%)
SUPPORT2	Mortality	Random Forest	–	0.884	0.676 (76.4%)	0.853 (96.5%)	0.610 (68.9%)	0.671 (75.8%)
SUPPORT2	Mortality	Gradient Boosting	–	0.902	0.671 (74.4%)	0.860 (95.3%)	0.636 (70.5%)	0.660 (73.1%)
SUPPORT2	Mortality	XGBoost	–	0.893	0.583 (65.3%)	0.835 (93.5%)	0.556 (62.2%)	0.476 (53.3%)
SUPPORT2	Mortality	MLP	–	0.893	0.674 (75.5%)	0.853 (95.5%)	0.715 (80.0%)	0.713 (79.8%)
UCI Cervical Cancer	Cervical Cancer	Logistic Regression	–	0.095	0.077 (81.9%)	0.074 (77.9%)	0.132 (139.1%)	0.082 (86.8%)
UCI Cervical Cancer	Cervical Cancer	Random Forest	–	0.194	0.091 (46.8%)	0.086 (44.4%)	0.123 (63.3%)	0.079 (40.5%)
UCI Cervical Cancer	Cervical Cancer	Gradient Boosting	–	0.138	0.089 (64.4%)	0.072 (52.4%)	0.106 (76.5%)	0.092 (67.0%)
UCI Cervical Cancer	Cervical Cancer	XGBoost	–	0.141	0.072 (51.1%)	0.157 (111.5%)	0.113 (80.4%)	0.063 (44.6%)
UCI Cervical Cancer	Cervical Cancer	MLP	–	0.059	0.107 (180.2%)	0.067 (112.0%)	0.055 (92.8%)	0.060 (100.7%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	Logistic Regression	–	1.000	0.966 (96.6%)	1.000 (100.0%)	0.980 (98.0%)	0.965 (96.5%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	Random Forest	–	1.000	0.986 (98.6%)	1.000 (100.0%)	0.994 (99.4%)	0.971 (97.1%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	Gradient Boosting	–	1.000	0.987 (98.7%)	0.999 (99.9%)	0.987 (98.7%)	0.863 (86.3%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	XGBoost	–	0.998	0.971 (97.3%)	1.000 (100.2%)	0.982 (98.4%)	0.879 (88.0%)
UCI Chronic Kidney Disease	Chronic Kidney Disease	MLP	–	0.992	0.982 (99.0%)	0.997 (100.4%)	0.960 (96.8%)	0.994 (100.1%)

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Table 6 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
UCI Dermatology	Dermatological Disease Classification	Logistic Regression	–	0.997	0.629 (63.1%)	0.978 (98.2%)	0.508 (51.0%)	0.513 (51.5%)
UCI Dermatology	Dermatological Disease Classification	Random Forest	–	0.991	0.604 (61.0%)	0.998 (100.7%)	0.591 (59.6%)	0.472 (47.7%)
UCI Dermatology	Dermatological Disease Classification	Gradient Boosting	–	0.976	0.607 (62.2%)	0.984 (100.8%)	0.596 (61.1%)	0.447 (45.8%)
UCI Dermatology	Dermatological Disease Classification	XGBoost	–	0.974	0.599 (61.5%)	0.993 (101.9%)	0.578 (59.3%)	0.387 (39.7%)
UCI Dermatology	Dermatological Disease Classification	MLP	–	0.973	0.527 (54.1%)	0.984 (101.2%)	0.607 (62.3%)	0.350 (36.0%)
UCI Diabetes 130-US Hospitals	30-day Readmission	Logistic Regression	–	0.203	0.116 (57.2%)	0.171 (83.9%)	0.135 (66.5%)	0.194 (95.4%)
UCI Diabetes 130-US Hospitals	30-day Readmission	Random Forest	–	0.218	0.113 (51.9%)	0.146 (67.2%)	0.127 (58.4%)	0.150 (68.8%)
UCI Diabetes 130-US Hospitals	30-day Readmission	Gradient Boosting	–	0.227	0.115 (50.9%)	0.126 (55.6%)	0.150 (66.1%)	0.147 (65.0%)
UCI Diabetes 130-US Hospitals	30-day Readmission	XGBoost	–	0.219	0.112 (51.2%)	0.159 (72.5%)	0.119 (54.6%)	0.133 (61.0%)
UCI Diabetes 130-US Hospitals	30-day Readmission	MLP	–	0.211	0.115 (54.3%)	0.179 (85.1%)	0.131 (62.1%)	0.146 (69.3%)
UCI Diabetes 130-US Hospitals	Any-Readmission	Logistic Regression	–	0.632	0.588 (93.0%)	0.621 (98.2%)	0.579 (91.5%)	0.619 (97.8%)
UCI Diabetes 130-US Hospitals	Any-Readmission	Random Forest	–	0.657	0.547 (83.2%)	0.600 (91.4%)	0.536 (81.6%)	0.587 (89.4%)
UCI Diabetes 130-US Hospitals	Any-Readmission	Gradient Boosting	–	0.653	0.574 (87.8%)	0.596 (91.2%)	0.564 (86.3%)	0.598 (91.5%)
UCI Diabetes 130-US Hospitals	Any-Readmission	XGBoost	–	0.668	0.558 (83.6%)	0.608 (91.0%)	0.541 (81.0%)	0.583 (87.3%)
UCI Diabetes 130-US Hospitals	Any-Readmission	MLP	–	0.653	0.567 (86.8%)	0.621 (95.1%)	0.573 (87.8%)	0.616 (94.3%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Logistic Regression	–	0.445	0.382 (85.8%)	0.416 (93.3%)	0.382 (85.8%)	0.401 (89.9%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Random Forest	–	0.467	0.380 (81.4%)	0.415 (88.9%)	0.381 (81.6%)	0.410 (87.8%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Gradient Boosting	–	0.474	0.396 (83.4%)	0.419 (88.2%)	0.373 (78.6%)	0.418 (88.2%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	XGBoost	–	0.487	0.381 (78.2%)	0.422 (86.7%)	0.380 (78.1%)	0.407 (83.5%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	MLP	–	0.466	0.382 (81.8%)	0.416 (89.3%)	0.386 (82.7%)	0.410 (88.0%)
UCI Early-Stage Diabetes	Early Diabetes Risk	Logistic Regression	–	0.994	0.994 (100.0%)	0.996 (100.2%)	0.983 (98.8%)	0.980 (98.5%)
UCI Early-Stage Diabetes	Early Diabetes Risk	Random Forest	–	0.999	0.992 (99.3%)	0.995 (99.7%)	0.978 (97.9%)	0.942 (94.3%)
UCI Early-Stage Diabetes	Early Diabetes Risk	Gradient Boosting	–	0.997	0.995 (99.8%)	0.992 (99.5%)	0.980 (98.3%)	0.956 (95.9%)
UCI Early-Stage Diabetes	Early Diabetes Risk	XGBoost	–	1.000	0.992 (99.2%)	0.992 (99.2%)	0.976 (97.6%)	0.943 (94.3%)
UCI Early-Stage Diabetes	Early Diabetes Risk	MLP	–	0.994	0.992 (99.8%)	0.960 (96.6%)	0.973 (98.0%)	0.946 (95.2%)
UCI Heart Failure	Heart Failure Mortality	Logistic Regression	0.495	0.591	0.418 (70.8%)	0.573 (96.9%)	0.740 (125.1%)	0.657 (111.1%)
UCI Heart Failure	Heart Failure Mortality	Random Forest	0.657	0.580	0.491 (84.6%)	0.745 (128.5%)	0.735 (126.7%)	0.565 (97.4%)
UCI Heart Failure	Heart Failure Mortality	Gradient Boosting	0.594	0.548	0.513 (93.6%)	0.738 (134.6%)	0.702 (128.2%)	0.459 (83.8%)
UCI Heart Failure	Heart Failure Mortality	XGBoost	0.594	0.569	0.539 (94.9%)	0.725 (127.6%)	0.606 (106.6%)	0.552 (97.1%)
UCI Heart Failure	Heart Failure Mortality	MLP	0.750	0.707	0.605 (85.6%)	0.685 (96.9%)	0.565 (79.9%)	0.623 (88.2%)
UCI Hepatitis	Hepatitis Mortality	Logistic Regression	–	0.968	0.963 (99.5%)	0.960 (99.2%)	0.866 (89.4%)	0.715 (73.9%)
UCI Hepatitis	Hepatitis Mortality	Random Forest	–	0.977	0.976 (99.8%)	0.981 (100.3%)	0.965 (98.7%)	0.950 (97.3%)
UCI Hepatitis	Hepatitis Mortality	Gradient Boosting	–	0.981	0.965 (98.4%)	0.965 (98.4%)	0.925 (94.3%)	0.963 (98.2%)
UCI Hepatitis	Hepatitis Mortality	XGBoost	–	0.974	0.986 (101.2%)	0.977 (100.4%)	0.915 (94.0%)	0.945 (97.0%)
UCI Hepatitis	Hepatitis Mortality	MLP	–	0.958	0.947 (99.0%)	0.962 (100.5%)	0.940 (98.2%)	0.859 (89.7%)
UCI Indian Liver Patient	Liver Disease	Logistic Regression	–	0.517	0.448 (86.8%)	0.489 (94.6%)	0.381 (73.7%)	0.536 (103.7%)
UCI Indian Liver Patient	Liver Disease	Random Forest	–	0.410	0.463 (112.9%)	0.467 (113.8%)	0.396 (96.4%)	0.388 (94.5%)
UCI Indian Liver Patient	Liver Disease	Gradient Boosting	–	0.357	0.369 (103.2%)	0.456 (127.5%)	0.419 (117.2%)	0.414 (115.9%)
UCI Indian Liver Patient	Liver Disease	XGBoost	–	0.382	0.368 (96.4%)	0.468 (122.7%)	0.382 (100.0%)	0.327 (85.5%)
UCI Indian Liver Patient	Liver Disease	MLP	–	0.514	0.490 (95.3%)	0.480 (93.4%)	0.440 (85.6%)	0.510 (99.3%)
UCI Mammographic Mass	Mass Malignancy	Logistic Regression	–	0.882	0.851 (96.5%)	0.893 (101.3%)	0.847 (96.0%)	0.769 (87.2%)
UCI Mammographic Mass	Mass Malignancy	Random Forest	–	0.801	0.826 (103.0%)	0.831 (103.8%)	0.781 (97.5%)	0.523 (65.2%)
UCI Mammographic Mass	Mass Malignancy	Gradient Boosting	–	0.833	0.891 (107.0%)	0.891 (107.1%)	0.848 (101.8%)	0.678 (81.5%)
UCI Mammographic Mass	Mass Malignancy	XGBoost	–	0.786	0.872 (110.9%)	0.877 (111.6%)	0.842 (107.1%)	0.540 (68.7%)
UCI Mammographic Mass	Mass Malignancy	MLP	–	0.875	0.859 (98.2%)	0.886 (101.3%)	0.853 (97.6%)	0.785 (89.8%)

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Table 6 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
UCI Statlog Heart	Heart Disease	Logistic Regression	–	0.865	0.823 (95.2%)	0.919 (106.3%)	0.850 (98.3%)	0.525 (60.7%)
UCI Statlog Heart	Heart Disease	Random Forest	–	0.810	0.802 (99.1%)	0.867 (107.1%)	0.762 (94.1%)	0.778 (96.1%)
UCI Statlog Heart	Heart Disease	Gradient Boosting	–	0.858	0.791 (92.2%)	0.879 (102.5%)	0.741 (86.4%)	0.701 (81.8%)
UCI Statlog Heart	Heart Disease	XGBoost	–	0.803	0.824 (102.6%)	0.906 (112.9%)	0.772 (96.1%)	0.749 (93.2%)
UCI Statlog Heart	Heart Disease	MLP	–	0.875	0.884 (101.0%)	0.867 (99.0%)	0.850 (97.2%)	0.444 (50.7%)
NHANES	Mortality	Logistic Regression	–	0.761	0.263 (34.5%)	0.704 (92.5%)	0.466 (61.3%)	0.709 (93.1%)
NHANES	Mortality	Random Forest	–	0.748	0.188 (25.1%)	0.720 (96.2%)	0.367 (49.1%)	0.660 (88.2%)
NHANES	Mortality	Gradient Boosting	–	0.773	0.261 (33.8%)	0.731 (94.7%)	0.404 (52.2%)	0.712 (92.2%)
NHANES	Mortality	XGBoost	–	0.780	0.183 (23.5%)	0.751 (96.2%)	0.380 (48.7%)	0.608 (77.9%)
NHANES	Mortality	MLP	–	0.747	0.225 (30.1%)	0.504 (67.5%)	0.430 (57.5%)	0.600 (80.3%)

4 Downstream Utility: Time-Series (Per-Classifier)

Table 7: Per-dataset, per-classifier time-series (ICU) downstream utility, AUROC (with the clinical-validity layer).

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
MIMIC-III	Heart Failure Mortality	Logistic Regression	0.842	0.777	0.533 (68.5%)	0.816 (105.0%)	0.556 (71.5%)	0.824 (106.0%)
MIMIC-III	Heart Failure Mortality	Random Forest	–	0.858	0.562 (65.5%)	0.852 (99.3%)	0.462 (53.9%)	0.763 (89.0%)
MIMIC-III	Heart Failure Mortality	Gradient Boosting	–	0.802	0.339 (42.3%)	0.785 (97.9%)	0.406 (50.6%)	0.717 (89.4%)
MIMIC-III	Heart Failure Mortality	XGBoost	–	0.815	0.578 (71.0%)	0.810 (99.5%)	0.466 (57.3%)	0.719 (88.3%)
MIMIC-III	Heart Failure Mortality	MLP	–	0.472	0.388 (82.2%)	0.787 (166.7%)	0.417 (88.4%)	0.753 (159.4%)
MIMIC-III	Hospital Readmission	Logistic Regression	0.580	0.611	0.533 (87.3%)	0.608 (99.6%)	0.544 (89.1%)	0.600 (98.3%)
MIMIC-III	Hospital Readmission	Random Forest	0.680	0.616	0.522 (84.9%)	0.622 (101.0%)	0.499 (81.1%)	0.484 (78.6%)
MIMIC-III	Hospital Readmission	Gradient Boosting	0.720	0.606	0.545 (90.0%)	0.602 (99.4%)	0.533 (88.1%)	0.531 (87.7%)
MIMIC-III	Hospital Readmission	XGBoost	0.740	0.575	0.480 (83.4%)	0.566 (98.3%)	0.500 (86.9%)	0.475 (82.5%)
MIMIC-III	Hospital Readmission	MLP	–	0.591	0.482 (81.6%)	0.582 (98.5%)	0.511 (86.6%)	0.473 (80.1%)
MIMIC-III	Length-of-Stay	Logistic Regression	–	0.739	0.648 (87.7%)	0.668 (90.5%)	0.649 (87.9%)	0.699 (94.7%)
MIMIC-III	Length-of-Stay	Random Forest	–	0.765	0.637 (83.3%)	0.715 (93.5%)	0.677 (88.5%)	0.575 (75.2%)
MIMIC-III	Length-of-Stay	Gradient Boosting	–	0.772	0.636 (82.4%)	0.729 (94.3%)	0.681 (88.2%)	0.649 (84.0%)
MIMIC-III	Length-of-Stay	XGBoost	–	0.764	0.558 (73.0%)	0.698 (91.3%)	0.608 (79.6%)	0.538 (70.4%)
MIMIC-III	Length-of-Stay	MLP	–	0.746	0.592 (79.4%)	0.684 (91.6%)	0.620 (83.1%)	0.599 (80.2%)
MIMIC-III	ICU Readmission	Logistic Regression	0.570	0.614	0.561 (91.5%)	0.600 (97.8%)	0.513 (83.7%)	0.552 (90.0%)
MIMIC-III	ICU Readmission	Random Forest	0.600	0.612	0.536 (87.7%)	0.606 (99.1%)	0.501 (81.9%)	0.517 (84.5%)
MIMIC-III	ICU Readmission	Gradient Boosting	0.620	0.612	0.543 (88.8%)	0.601 (98.2%)	0.519 (84.9%)	0.541 (88.5%)
MIMIC-III	ICU Readmission	XGBoost	0.600	0.577	0.475 (82.3%)	0.574 (99.5%)	0.484 (83.8%)	0.483 (83.7%)
MIMIC-III	ICU Readmission	MLP	–	0.591	0.494 (83.5%)	0.581 (98.2%)	0.470 (79.6%)	0.468 (79.1%)
MIMIC-III	COPD Mortality	Logistic Regression	–	0.697	0.444 (63.6%)	0.656 (94.1%)	0.517 (74.2%)	0.573 (82.2%)
MIMIC-III	COPD Mortality	Random Forest	–	0.728	0.392 (53.8%)	0.670 (92.1%)	0.449 (61.7%)	0.523 (71.9%)
MIMIC-III	COPD Mortality	Gradient Boosting	–	0.704	0.428 (60.9%)	0.628 (89.2%)	0.408 (58.0%)	0.495 (70.3%)
MIMIC-III	COPD Mortality	XGBoost	–	0.713	0.425 (59.6%)	0.632 (88.6%)	0.459 (64.3%)	0.552 (77.5%)
MIMIC-III	COPD Mortality	MLP	–	0.716	0.373 (52.1%)	0.594 (83.0%)	0.389 (54.4%)	0.542 (75.7%)

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Table 7 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
MIMIC-III	Hospital Mortality	Logistic Regression	–	0.820	0.700 (85.4%)	0.750 (91.5%)	0.670 (81.7%)	0.640 (78.0%)
MIMIC-III	Hospital Mortality	Random Forest	–	0.830	0.690 (83.1%)	0.770 (92.8%)	0.670 (80.7%)	0.620 (74.7%)
MIMIC-III	Hospital Mortality	Gradient Boosting	–	0.840	0.710 (84.5%)	0.780 (92.9%)	0.680 (81.0%)	0.650 (77.4%)
MIMIC-III	Hospital Mortality	MLP	–	0.850	0.660 (77.6%)	0.790 (92.9%)	0.640 (75.3%)	0.580 (68.2%)
MIMIC-IV	COPD Mortality	Logistic Regression	–	0.833	0.590 (70.9%)	0.772 (92.7%)	0.569 (68.3%)	0.663 (79.6%)
MIMIC-IV	COPD Mortality	Random Forest	–	0.816	0.564 (69.1%)	0.768 (94.1%)	0.573 (70.2%)	0.585 (71.6%)
MIMIC-IV	COPD Mortality	Gradient Boosting	–	0.829	0.569 (68.6%)	0.773 (93.3%)	0.524 (63.2%)	0.545 (65.8%)
MIMIC-IV	COPD Mortality	XGBoost	–	0.812	0.527 (64.8%)	0.775 (95.4%)	0.497 (61.1%)	0.531 (65.3%)
MIMIC-IV	COPD Mortality	MLP	–	0.847	0.530 (62.6%)	0.761 (89.9%)	0.565 (66.7%)	0.462 (54.5%)
MIMIC-IV	Heart Failure Mortality	Logistic Regression	–	0.821	0.642 (78.2%)	0.757 (92.2%)	0.696 (84.7%)	0.728 (88.7%)
MIMIC-IV	Heart Failure Mortality	Random Forest	–	0.829	0.574 (69.2%)	0.773 (93.3%)	0.649 (78.3%)	0.582 (70.2%)
MIMIC-IV	Heart Failure Mortality	Gradient Boosting	–	0.839	0.535 (63.8%)	0.786 (93.7%)	0.657 (78.3%)	0.588 (70.1%)
MIMIC-IV	Heart Failure Mortality	XGBoost	0.820	0.828	0.474 (57.3%)	0.780 (94.2%)	0.542 (65.5%)	0.494 (59.6%)
MIMIC-IV	Heart Failure Mortality	MLP	–	0.815	0.549 (67.4%)	0.742 (91.1%)	0.615 (75.5%)	0.413 (50.6%)
MIMIC-IV	Hospital Readmission	Logistic Regression	0.675	0.620	0.528 (85.2%)	0.510 (82.4%)	0.491 (79.3%)	0.492 (79.4%)
MIMIC-IV	Hospital Readmission	Random Forest	–	0.632	0.515 (81.5%)	0.521 (82.4%)	0.510 (80.7%)	0.512 (81.0%)
MIMIC-IV	Hospital Readmission	Gradient Boosting	–	0.627	0.519 (82.8%)	0.520 (83.0%)	0.503 (80.2%)	0.531 (84.6%)
MIMIC-IV	Hospital Readmission	XGBoost	0.696	0.624	0.506 (81.1%)	0.517 (82.9%)	0.515 (82.5%)	0.516 (82.8%)
MIMIC-IV	Hospital Readmission	MLP	–	0.607	0.528 (87.1%)	0.503 (82.9%)	0.490 (80.8%)	0.503 (82.9%)
MIMIC-IV	Length-of-Stay	Logistic Regression	–	0.815	0.801 (98.3%)	0.756 (92.8%)	0.794 (97.5%)	0.779 (95.7%)
MIMIC-IV	Length-of-Stay	Random Forest	–	0.831	0.789 (95.0%)	0.776 (93.4%)	0.788 (94.8%)	0.740 (89.1%)
MIMIC-IV	Length-of-Stay	Gradient Boosting	–	0.829	0.808 (97.5%)	0.780 (94.1%)	0.806 (97.2%)	0.766 (92.3%)
MIMIC-IV	Length-of-Stay	XGBoost	0.860	0.835	0.782 (93.7%)	0.769 (92.1%)	0.775 (92.8%)	0.712 (85.3%)
MIMIC-IV	Length-of-Stay	MLP	–	0.831	0.791 (95.1%)	0.748 (89.9%)	0.779 (93.7%)	0.752 (90.5%)
MIMIC-IV	ICU Mortality	Logistic Regression	–	0.889	0.782 (88.0%)	0.854 (96.0%)	0.785 (88.2%)	0.810 (91.1%)
MIMIC-IV	ICU Mortality	Random Forest	–	0.892	0.773 (86.7%)	0.858 (96.2%)	0.764 (85.7%)	0.683 (76.7%)
MIMIC-IV	ICU Mortality	Gradient Boosting	–	0.897	0.799 (89.0%)	0.867 (96.6%)	0.793 (88.5%)	0.769 (85.7%)
MIMIC-IV	ICU Mortality	XGBoost	0.870	0.897	0.634 (70.7%)	0.863 (96.2%)	0.649 (72.3%)	0.527 (58.7%)
MIMIC-IV	ICU Mortality	MLP	–	0.898	0.696 (77.5%)	0.861 (95.9%)	0.738 (82.1%)	0.588 (65.4%)
MIMIC-IV	ICU Readmission	Logistic Regression	0.570	0.650	0.500 (77.0%)	0.525 (80.9%)	0.507 (78.1%)	0.551 (84.8%)
MIMIC-IV	ICU Readmission	Random Forest	0.600	0.660	0.509 (77.2%)	0.659 (99.8%)	0.524 (79.3%)	0.493 (74.7%)
MIMIC-IV	ICU Readmission	Gradient Boosting	0.620	0.667	0.492 (73.8%)	0.641 (96.0%)	0.497 (74.5%)	0.532 (79.8%)
MIMIC-IV	ICU Readmission	XGBoost	0.600	0.614	0.459 (74.8%)	0.649 (105.7%)	0.516 (84.1%)	0.512 (83.4%)
MIMIC-IV	ICU Readmission	MLP	–	0.628	0.492 (78.3%)	0.573 (91.2%)	0.505 (80.3%)	0.504 (80.2%)
eICU	COPD Mortality	Logistic Regression	–	0.801	0.710 (88.7%)	0.763 (95.3%)	0.676 (84.4%)	0.638 (79.7%)
eICU	COPD Mortality	Random Forest	–	0.801	0.536 (66.9%)	0.745 (93.1%)	0.598 (74.7%)	0.604 (75.4%)
eICU	COPD Mortality	Gradient Boosting	–	0.800	0.491 (61.3%)	0.747 (93.3%)	0.548 (68.5%)	0.517 (64.6%)
eICU	COPD Mortality	XGBoost	–	0.780	0.509 (65.3%)	0.728 (93.3%)	0.535 (68.5%)	0.486 (62.3%)
eICU	COPD Mortality	MLP	–	0.787	0.536 (68.1%)	0.764 (97.1%)	0.621 (78.9%)	0.492 (62.5%)
eICU	Heart Failure Mortality	Logistic Regression	–	0.747	0.666 (89.1%)	0.733 (98.0%)	0.638 (85.3%)	0.737 (98.7%)
eICU	Heart Failure Mortality	Random Forest	–	0.753	0.633 (84.1%)	0.712 (94.5%)	0.622 (82.6%)	0.689 (91.4%)
eICU	Heart Failure Mortality	Gradient Boosting	–	0.750	0.616 (82.1%)	0.713 (95.1%)	0.592 (78.9%)	0.683 (91.1%)
eICU	Heart Failure Mortality	XGBoost	–	0.719	0.558 (77.7%)	0.702 (97.7%)	0.545 (75.9%)	0.585 (81.4%)
eICU	Heart Failure Mortality	MLP	–	0.724	0.586 (81.0%)	0.712 (98.3%)	0.593 (81.9%)	0.612 (84.6%)

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Table 7 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
eICU	Length-of-Stay	Logistic Regression	–	0.746	0.716 (96.0%)	0.725 (97.1%)	0.719 (96.3%)	0.709 (95.0%)
eICU	Length-of-Stay	Random Forest	–	0.762	0.697 (91.4%)	0.729 (95.6%)	0.699 (91.7%)	0.620 (81.4%)
eICU	Length-of-Stay	Gradient Boosting	–	0.763	0.719 (94.2%)	0.736 (96.5%)	0.720 (94.4%)	0.699 (91.6%)
eICU	Length-of-Stay	XGBoost	–	0.747	0.636 (85.1%)	0.719 (96.3%)	0.638 (85.3%)	0.573 (76.6%)
eICU	Length-of-Stay	MLP	–	0.757	0.693 (91.5%)	0.706 (93.2%)	0.696 (91.9%)	0.641 (84.6%)
eICU	Mortality	Logistic Regression	–	0.888	0.879 (99.0%)	0.802 (90.3%)	0.878 (98.9%)	0.850 (95.7%)
eICU	Mortality	Random Forest	–	0.900	0.875 (97.2%)	0.855 (95.0%)	0.870 (96.7%)	0.723 (80.3%)
eICU	Mortality	Gradient Boosting	–	0.901	0.887 (98.4%)	0.868 (96.3%)	0.882 (97.8%)	0.821 (91.0%)
eICU	Mortality	XGBoost	–	0.903	0.847 (93.8%)	0.874 (96.7%)	0.852 (94.3%)	0.557 (61.7%)
eICU	Mortality	MLP	–	0.899	0.863 (96.0%)	0.845 (94.0%)	0.866 (96.3%)	0.733 (81.6%)
eICU	Readmission	Logistic Regression	–	0.665	0.585 (87.9%)	0.589 (88.6%)	0.562 (84.4%)	0.585 (87.8%)
eICU	Readmission	Random Forest	0.680	0.706	0.597 (84.5%)	0.619 (87.7%)	0.578 (81.8%)	0.489 (69.3%)
eICU	Readmission	Gradient Boosting	–	0.707	0.632 (89.3%)	0.621 (87.9%)	0.612 (86.5%)	0.557 (78.8%)
eICU	Readmission	XGBoost	–	0.718	0.575 (80.2%)	0.615 (85.7%)	0.569 (79.3%)	0.506 (70.6%)
eICU	Readmission	MLP	–	0.679	0.584 (86.0%)	0.603 (88.8%)	0.559 (82.3%)	0.464 (68.3%)

Table 8: Per-dataset, per-classifier time-series (ICU) downstream utility, AUPRC (with the clinical-validity layer).

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
MIMIC-III	Heart Failure Mortality	Logistic Regression	–	0.413	0.136 (32.9%)	0.459 (111.1%)	0.180 (43.5%)	0.411 (99.4%)
MIMIC-III	Heart Failure Mortality	Random Forest	–	0.548	0.136 (24.9%)	0.493 (90.0%)	0.110 (20.1%)	0.336 (61.4%)
MIMIC-III	Heart Failure Mortality	Gradient Boosting	–	0.509	0.089 (17.4%)	0.415 (81.6%)	0.102 (20.1%)	0.316 (62.1%)
MIMIC-III	Heart Failure Mortality	XGBoost	–	0.475	0.149 (31.4%)	0.465 (97.9%)	0.114 (24.0%)	0.290 (61.1%)
MIMIC-III	Heart Failure Mortality	MLP	–	0.122	0.123 (100.6%)	0.435 (355.8%)	0.110 (89.7%)	0.325 (265.5%)
MIMIC-III	Hospital Readmission	Logistic Regression	–	0.078	0.063 (80.6%)	0.077 (98.9%)	0.064 (82.8%)	0.070 (90.6%)
MIMIC-III	Hospital Readmission	Random Forest	–	0.079	0.056 (70.9%)	0.082 (104.2%)	0.051 (64.2%)	0.054 (68.6%)
MIMIC-III	Hospital Readmission	Gradient Boosting	–	0.077	0.065 (84.5%)	0.079 (103.0%)	0.060 (77.5%)	0.058 (75.4%)
MIMIC-III	Hospital Readmission	XGBoost	–	0.072	0.053 (72.8%)	0.079 (109.1%)	0.056 (77.5%)	0.055 (76.7%)
MIMIC-III	Hospital Readmission	MLP	–	0.073	0.052 (70.9%)	0.075 (103.4%)	0.056 (76.6%)	0.050 (68.3%)
MIMIC-III	Length-of-Stay	Logistic Regression	–	0.340	0.254 (74.8%)	0.268 (78.6%)	0.244 (71.9%)	0.298 (87.5%)
MIMIC-III	Length-of-Stay	Random Forest	–	0.376	0.237 (63.0%)	0.292 (77.7%)	0.267 (71.0%)	0.187 (49.9%)
MIMIC-III	Length-of-Stay	Gradient Boosting	–	0.387	0.259 (67.0%)	0.317 (81.9%)	0.280 (72.3%)	0.269 (69.6%)
MIMIC-III	Length-of-Stay	XGBoost	–	0.396	0.212 (53.7%)	0.277 (69.9%)	0.244 (61.7%)	0.193 (48.7%)
MIMIC-III	Length-of-Stay	MLP	–	0.370	0.227 (61.4%)	0.292 (78.9%)	0.246 (66.5%)	0.249 (67.3%)
MIMIC-III	ICU Readmission	Logistic Regression	–	0.149	0.123 (82.7%)	0.134 (90.0%)	0.105 (70.6%)	0.118 (79.4%)
MIMIC-III	ICU Readmission	Random Forest	–	0.134	0.108 (80.9%)	0.139 (104.2%)	0.096 (72.0%)	0.104 (77.8%)
MIMIC-III	ICU Readmission	Gradient Boosting	–	0.143	0.116 (81.2%)	0.133 (93.1%)	0.104 (72.8%)	0.116 (80.6%)
MIMIC-III	ICU Readmission	XGBoost	–	0.128	0.093 (72.6%)	0.132 (103.3%)	0.095 (74.0%)	0.095 (74.6%)
MIMIC-III	ICU Readmission	MLP	–	0.131	0.097 (74.1%)	0.139 (106.0%)	0.094 (71.8%)	0.093 (71.1%)
MIMIC-III	COPD Mortality	Logistic Regression	–	0.296	0.146 (49.4%)	0.273 (92.0%)	0.168 (56.6%)	0.225 (75.9%)
MIMIC-III	COPD Mortality	Random Forest	–	0.355	0.113 (31.8%)	0.237 (66.7%)	0.123 (34.5%)	0.170 (47.9%)
MIMIC-III	COPD Mortality	Gradient Boosting	–	0.334	0.124 (37.0%)	0.209 (62.6%)	0.117 (34.9%)	0.164 (49.0%)
MIMIC-III	COPD Mortality	XGBoost	–	0.353	0.122 (34.4%)	0.203 (57.6%)	0.134 (37.8%)	0.170 (48.1%)
MIMIC-III	COPD Mortality	MLP	–	0.376	0.108 (28.8%)	0.210 (55.8%)	0.114 (30.2%)	0.220 (58.5%)

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Table 8 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
MIMIC-III	Hospital Mortality	Logistic Regression	–	0.440	0.290 (65.9%)	0.340 (77.3%)	0.260 (59.1%)	0.230 (52.3%)
MIMIC-III	Hospital Mortality	Random Forest	–	0.460	0.270 (58.7%)	0.350 (76.1%)	0.250 (54.3%)	0.200 (43.5%)
MIMIC-III	Hospital Mortality	Gradient Boosting	–	0.480	0.300 (62.5%)	0.380 (79.2%)	0.270 (56.3%)	0.240 (50.0%)
MIMIC-III	Hospital Mortality	MLP	–	0.490	0.220 (44.9%)	0.360 (73.5%)	0.200 (40.8%)	0.170 (34.7%)
MIMIC-IV	COPD Mortality	Logistic Regression	–	0.527	0.268 (51.0%)	0.389 (73.8%)	0.257 (48.8%)	0.312 (59.3%)
MIMIC-IV	COPD Mortality	Random Forest	–	0.488	0.203 (41.5%)	0.415 (84.9%)	0.207 (42.5%)	0.261 (53.3%)
MIMIC-IV	COPD Mortality	Gradient Boosting	–	0.495	0.234 (47.3%)	0.411 (83.1%)	0.228 (46.0%)	0.209 (42.2%)
MIMIC-IV	COPD Mortality	XGBoost	–	0.485	0.205 (42.3%)	0.411 (84.8%)	0.188 (38.8%)	0.208 (42.9%)
MIMIC-IV	COPD Mortality	MLP	–	0.565	0.192 (34.0%)	0.399 (70.5%)	0.262 (46.3%)	0.211 (37.3%)
MIMIC-IV	Heart Failure Mortality	Logistic Regression	–	0.471	0.255 (54.2%)	0.356 (75.6%)	0.289 (61.5%)	0.344 (73.1%)
MIMIC-IV	Heart Failure Mortality	Random Forest	–	0.460	0.162 (35.2%)	0.365 (79.3%)	0.202 (43.9%)	0.169 (36.6%)
MIMIC-IV	Heart Failure Mortality	Gradient Boosting	–	0.492	0.166 (33.8%)	0.390 (79.4%)	0.245 (49.9%)	0.231 (47.1%)
MIMIC-IV	Heart Failure Mortality	XGBoost	–	0.482	0.143 (29.6%)	0.389 (80.7%)	0.181 (37.7%)	0.154 (31.9%)
MIMIC-IV	Heart Failure Mortality	MLP	–	0.466	0.177 (38.0%)	0.362 (77.8%)	0.231 (49.7%)	0.146 (31.3%)
MIMIC-IV	Hospital Readmission	Logistic Regression	–	0.346	0.277 (80.0%)	0.256 (74.0%)	0.251 (72.7%)	0.252 (72.8%)
MIMIC-IV	Hospital Readmission	Random Forest	–	0.379	0.256 (67.7%)	0.278 (73.5%)	0.265 (70.0%)	0.267 (70.4%)
MIMIC-IV	Hospital Readmission	Gradient Boosting	–	0.362	0.258 (71.3%)	0.280 (77.4%)	0.257 (70.9%)	0.282 (77.8%)
MIMIC-IV	Hospital Readmission	XGBoost	–	0.363	0.262 (72.0%)	0.273 (75.0%)	0.268 (73.6%)	0.267 (73.3%)
MIMIC-IV	Hospital Readmission	MLP	–	0.345	0.277 (80.2%)	0.257 (74.4%)	0.248 (72.1%)	0.258 (74.9%)
MIMIC-IV	Length-of-Stay	Logistic Regression	–	0.628	0.613 (97.5%)	0.562 (89.5%)	0.612 (97.5%)	0.592 (94.2%)
MIMIC-IV	Length-of-Stay	Random Forest	–	0.669	0.586 (87.7%)	0.582 (87.0%)	0.590 (88.3%)	0.526 (78.7%)
MIMIC-IV	Length-of-Stay	Gradient Boosting	–	0.670	0.630 (94.0%)	0.599 (89.4%)	0.633 (94.5%)	0.579 (86.3%)
MIMIC-IV	Length-of-Stay	XGBoost	0.750	0.693	0.594 (85.7%)	0.589 (85.0%)	0.596 (86.1%)	0.519 (75.0%)
MIMIC-IV	Length-of-Stay	MLP	–	0.681	0.605 (88.7%)	0.553 (81.1%)	0.594 (87.2%)	0.563 (82.6%)
MIMIC-IV	ICU Mortality	Logistic Regression	–	0.519	0.265 (51.0%)	0.365 (70.2%)	0.268 (51.6%)	0.322 (62.0%)
MIMIC-IV	ICU Mortality	Random Forest	–	0.511	0.237 (46.4%)	0.416 (81.4%)	0.242 (47.4%)	0.184 (36.1%)
MIMIC-IV	ICU Mortality	Gradient Boosting	–	0.533	0.309 (57.9%)	0.442 (83.1%)	0.306 (57.4%)	0.331 (62.2%)
MIMIC-IV	ICU Mortality	XGBoost	0.520	0.552	0.174 (31.5%)	0.435 (78.8%)	0.195 (35.4%)	0.137 (24.8%)
MIMIC-IV	ICU Mortality	MLP	–	0.551	0.212 (38.4%)	0.377 (68.4%)	0.247 (44.8%)	0.201 (36.5%)
MIMIC-IV	ICU Readmission	Logistic Regression	–	0.312	0.198 (63.4%)	0.196 (62.9%)	0.194 (62.0%)	0.221 (70.7%)
MIMIC-IV	ICU Readmission	Random Forest	–	0.319	0.194 (60.9%)	0.286 (89.5%)	0.203 (63.6%)	0.191 (59.9%)
MIMIC-IV	ICU Readmission	Gradient Boosting	–	0.309	0.202 (65.4%)	0.303 (98.1%)	0.199 (64.4%)	0.210 (67.9%)
MIMIC-IV	ICU Readmission	XGBoost	–	0.273	0.178 (65.5%)	0.306 (112.2%)	0.192 (70.4%)	0.194 (71.0%)
MIMIC-IV	ICU Readmission	MLP	–	0.283	0.184 (64.9%)	0.233 (82.2%)	0.190 (67.0%)	0.190 (67.0%)
eICU	COPD Mortality	Logistic Regression	–	0.368	0.260 (70.6%)	0.300 (81.5%)	0.217 (58.9%)	0.210 (56.9%)
eICU	COPD Mortality	Random Forest	–	0.378	0.121 (31.9%)	0.257 (68.1%)	0.134 (35.4%)	0.161 (42.5%)
eICU	COPD Mortality	Gradient Boosting	–	0.360	0.127 (35.2%)	0.281 (78.1%)	0.118 (32.7%)	0.164 (45.5%)
eICU	COPD Mortality	XGBoost	–	0.353	0.118 (33.5%)	0.265 (75.2%)	0.116 (32.7%)	0.114 (32.3%)
eICU	COPD Mortality	MLP	–	0.365	0.158 (43.2%)	0.296 (81.2%)	0.179 (49.1%)	0.122 (33.5%)
eICU	Heart Failure Mortality	Logistic Regression	–	0.330	0.244 (74.1%)	0.323 (98.2%)	0.220 (66.9%)	0.332 (100.9%)
eICU	Heart Failure Mortality	Random Forest	–	0.328	0.191 (58.3%)	0.254 (77.6%)	0.177 (53.9%)	0.245 (74.9%)
eICU	Heart Failure Mortality	Gradient Boosting	–	0.347	0.195 (56.2%)	0.282 (81.2%)	0.184 (52.9%)	0.252 (72.5%)
eICU	Heart Failure Mortality	XGBoost	–	0.305	0.156 (51.1%)	0.273 (89.5%)	0.158 (51.7%)	0.185 (60.7%)
eICU	Heart Failure Mortality	MLP	–	0.325	0.185 (56.8%)	0.295 (90.8%)	0.190 (58.5%)	0.217 (66.6%)

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Table 8 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
eICU	Length-of-Stay	Logistic Regression	–	0.270	0.238 (88.2%)	0.240 (88.8%)	0.234 (86.8%)	0.241 (89.1%)
eICU	Length-of-Stay	Random Forest	–	0.282	0.204 (72.6%)	0.242 (86.0%)	0.215 (76.3%)	0.152 (54.1%)
eICU	Length-of-Stay	Gradient Boosting	–	0.301	0.243 (80.7%)	0.252 (83.8%)	0.245 (81.4%)	0.243 (80.9%)
eICU	Length-of-Stay	XGBoost	–	0.279	0.185 (66.5%)	0.241 (86.6%)	0.190 (68.1%)	0.157 (56.2%)
eICU	Length-of-Stay	MLP	–	0.290	0.221 (76.3%)	0.221 (76.3%)	0.229 (78.9%)	0.201 (69.2%)
eICU	Mortality	Logistic Regression	–	0.400	0.388 (97.2%)	0.287 (71.8%)	0.385 (96.4%)	0.353 (88.3%)
eICU	Mortality	Random Forest	–	0.427	0.356 (83.4%)	0.377 (88.4%)	0.355 (83.2%)	0.117 (27.3%)
eICU	Mortality	Gradient Boosting	–	0.459	0.406 (88.4%)	0.397 (86.5%)	0.400 (87.3%)	0.308 (67.2%)
eICU	Mortality	XGBoost	–	0.471	0.361 (76.5%)	0.407 (86.4%)	0.356 (75.5%)	0.111 (23.6%)
eICU	Mortality	MLP	–	0.432	0.389 (89.9%)	0.325 (75.0%)	0.381 (88.1%)	0.223 (51.5%)
eICU	Readmission	Logistic Regression	–	0.301	0.197 (65.6%)	0.205 (68.1%)	0.186 (61.8%)	0.213 (71.0%)
eICU	Readmission	Random Forest	–	0.370	0.199 (53.8%)	0.209 (56.5%)	0.189 (51.1%)	0.151 (40.8%)
eICU	Readmission	Gradient Boosting	–	0.380	0.238 (62.6%)	0.222 (58.5%)	0.225 (59.1%)	0.192 (50.5%)
eICU	Readmission	XGBoost	–	0.408	0.193 (47.3%)	0.216 (53.0%)	0.200 (48.9%)	0.167 (40.8%)
eICU	Readmission	MLP	–	0.331	0.196 (59.1%)	0.211 (63.6%)	0.183 (55.1%)	0.142 (42.9%)

5 Multi-class Classification Results

Three tasks in the benchmark are multi-class: SUPPORT2 Functional Outcome (5 classes), UCI Dermatology (6 classes), and UCI Diabetes 130 Readmission (3 classes). For these tasks AUROC is the macro one-vs-rest AUROC and AUPRC is the macro AUPRC, computed across all classes.

Table 9: Multi-class tabular downstream utility, macro one-vs-rest AUROC (per classifier).

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
SUPPORT2	Functional Outcome	Logistic Regression	–	0.807	0.555 (68.7%)	0.748 (92.7%)	0.555 (68.7%)	0.567 (70.3%)
SUPPORT2	Functional Outcome	Random Forest	–	0.821	0.607 (73.9%)	0.773 (94.2%)	0.598 (72.8%)	0.612 (74.5%)
SUPPORT2	Functional Outcome	Gradient Boosting	–	0.833	0.602 (72.3%)	0.780 (93.5%)	0.600 (72.0%)	0.603 (72.4%)
SUPPORT2	Functional Outcome	XGBoost	–	0.823	0.580 (70.5%)	0.774 (94.0%)	0.574 (69.7%)	0.589 (71.6%)
SUPPORT2	Functional Outcome	MLP	–	0.826	0.545 (66.0%)	0.782 (94.6%)	0.554 (67.0%)	0.568 (68.8%)
UCI Dermatology	Dermatological Disease Classification	Logistic Regression	–	0.999	0.834 (83.4%)	0.995 (99.5%)	0.853 (85.3%)	0.781 (78.2%)
UCI Dermatology	Dermatological Disease Classification	Random Forest	–	0.998	0.872 (87.4%)	1.000 (100.1%)	0.856 (85.7%)	0.779 (78.1%)
UCI Dermatology	Dermatological Disease Classification	Gradient Boosting	–	0.995	0.872 (87.7%)	0.997 (100.2%)	0.871 (87.6%)	0.709 (71.2%)
UCI Dermatology	Dermatological Disease Classification	XGBoost	–	0.995	0.874 (87.8%)	0.998 (100.3%)	0.852 (85.6%)	0.736 (73.9%)
UCI Dermatology	Dermatological Disease Classification	MLP	–	0.994	0.830 (83.5%)	0.996 (100.2%)	0.898 (90.4%)	0.693 (69.8%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Logistic Regression	–	0.646	0.561 (86.8%)	0.610 (94.3%)	0.570 (88.3%)	0.579 (89.7%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Random Forest	–	0.672	0.566 (84.3%)	0.617 (91.8%)	0.569 (84.7%)	0.611 (91.0%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Gradient Boosting	–	0.675	0.589 (87.2%)	0.614 (90.9%)	0.555 (82.2%)	0.621 (92.0%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	XGBoost	–	0.687	0.568 (82.7%)	0.625 (91.0%)	0.571 (83.1%)	0.611 (88.9%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	MLP	–	0.667	0.566 (84.8%)	0.607 (90.9%)	0.577 (86.5%)	0.615 (92.2%)

Table 10: Multi-class tabular downstream utility, macro AUPRC (per classifier).

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
SUPPORT2	Functional Outcome	Logistic Regression	–	0.449	0.287 (64.0%)	0.400 (89.2%)	0.298 (66.3%)	0.301 (67.0%)
SUPPORT2	Functional Outcome	Random Forest	–	0.454	0.275 (60.7%)	0.413 (91.2%)	0.310 (68.4%)	0.321 (70.7%)
SUPPORT2	Functional Outcome	Gradient Boosting	–	0.469	0.257 (54.7%)	0.410 (87.5%)	0.301 (64.1%)	0.312 (66.5%)
SUPPORT2	Functional Outcome	XGBoost	–	0.459	0.256 (55.8%)	0.399 (86.9%)	0.296 (64.6%)	0.303 (66.0%)
SUPPORT2	Functional Outcome	MLP	–	0.462	0.296 (64.1%)	0.410 (88.6%)	0.311 (67.3%)	0.315 (68.2%)
UCI Dermatology	Dermatological Disease Classification	Logistic Regression	–	0.997	0.629 (63.1%)	0.978 (98.2%)	0.508 (51.0%)	0.513 (51.5%)
UCI Dermatology	Dermatological Disease Classification	Random Forest	–	0.991	0.604 (61.0%)	0.998 (100.7%)	0.591 (59.6%)	0.472 (47.7%)
UCI Dermatology	Dermatological Disease Classification	Gradient Boosting	–	0.976	0.607 (62.2%)	0.984 (100.8%)	0.596 (61.1%)	0.447 (45.8%)
UCI Dermatology	Dermatological Disease Classification	XGBoost	–	0.974	0.599 (61.5%)	0.993 (101.9%)	0.578 (59.3%)	0.387 (39.7%)
UCI Dermatology	Dermatological Disease Classification	MLP	–	0.973	0.527 (54.1%)	0.984 (101.2%)	0.607 (62.3%)	0.350 (36.0%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Logistic Regression	–	0.445	0.382 (85.8%)	0.416 (93.3%)	0.382 (85.8%)	0.401 (89.9%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Random Forest	–	0.467	0.380 (81.4%)	0.415 (88.9%)	0.381 (81.6%)	0.410 (87.8%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	Gradient Boosting	–	0.474	0.396 (83.4%)	0.419 (88.2%)	0.373 (78.6%)	0.418 (88.2%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	XGBoost	–	0.487	0.381 (78.2%)	0.422 (86.7%)	0.380 (78.1%)	0.407 (83.5%)
UCI Diabetes 130-US Hospitals	Readmission (3-class)	MLP	–	0.466	0.382 (81.8%)	0.416 (89.3%)	0.386 (82.7%)	0.410 (88.0%)

6 Ablation: Effect of the Clinical-Validity Layer

To isolate the contribution of the model-agnostic clinical-validity layer, we re-run the utility evaluation with the layer disabled and report AUROC retention. Only datasets that carry a validity requirement are included; comparing against the main utility tables quantifies how much downstream signal is attributable to enforcing clinical and structural validity.

Table 11: Tabular ablation: downstream utility (AUROC, Retention) with the clinical-validity layer *disabled*.

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
SUPPORT2	Death	Logistic Regression	–	0.841	0.788 (93.8%)	0.820 (97.6%)	0.809 (96.2%)	0.792 (94.2%)
SUPPORT2	Death	Random Forest	–	0.849	0.783 (92.2%)	0.825 (97.2%)	0.814 (95.9%)	0.786 (92.6%)
SUPPORT2	Death	Gradient Boosting	–	0.855	0.789 (92.2%)	0.827 (96.6%)	0.809 (94.6%)	0.776 (90.7%)
SUPPORT2	Death	XGBoost	–	0.847	0.763 (90.0%)	0.828 (97.8%)	0.791 (93.4%)	0.763 (90.1%)
SUPPORT2	Death	MLP	–	0.846	0.726 (85.8%)	0.824 (97.3%)	0.801 (94.7%)	0.759 (89.7%)
SUPPORT2	Functional Outcome	Logistic Regression	–	0.807	0.762 (94.3%)	0.817 (101.2%)	0.706 (87.4%)	0.576 (71.3%)
SUPPORT2	Functional Outcome	Random Forest	–	0.821	0.757 (92.2%)	0.819 (99.8%)	0.767 (93.5%)	0.570 (69.5%)
SUPPORT2	Functional Outcome	Gradient Boosting	–	0.833	0.763 (91.5%)	0.824 (98.9%)	0.766 (91.9%)	0.599 (71.9%)
SUPPORT2	Functional Outcome	XGBoost	–	0.823	0.762 (92.6%)	0.838 (101.7%)	0.742 (90.2%)	0.593 (72.1%)
SUPPORT2	Functional Outcome	MLP	–	0.826	0.669 (81.0%)	0.826 (100.0%)	0.714 (86.5%)	0.557 (67.4%)
SUPPORT2	Mortality	Logistic Regression	–	0.946	0.929 (98.2%)	0.943 (99.7%)	0.929 (98.2%)	0.919 (97.1%)
SUPPORT2	Mortality	Random Forest	–	0.949	0.933 (98.3%)	0.942 (99.2%)	0.929 (98.0%)	0.879 (92.6%)
SUPPORT2	Mortality	Gradient Boosting	–	0.951	0.925 (97.3%)	0.944 (99.2%)	0.927 (97.5%)	0.854 (89.8%)
SUPPORT2	Mortality	XGBoost	–	0.950	0.910 (95.8%)	0.946 (99.6%)	0.918 (96.6%)	0.748 (78.7%)
SUPPORT2	Mortality	MLP	–	0.951	0.927 (97.6%)	0.942 (99.1%)	0.927 (97.5%)	0.883 (92.8%)

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Table 11 (continued)

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
UCI Cervical Cancer	Cervical Cancer	Logistic Regression	–	0.574	0.392 (68.3%)	0.473 (82.4%)	0.452 (78.7%)	0.385 (67.0%)
UCI Cervical Cancer	Cervical Cancer	Random Forest	–	0.698	0.577 (82.7%)	0.580 (83.1%)	0.411 (58.9%)	0.643 (92.1%)
UCI Cervical Cancer	Cervical Cancer	Gradient Boosting	–	0.665	0.430 (64.6%)	0.600 (90.2%)	0.403 (60.5%)	0.468 (70.3%)
UCI Cervical Cancer	Cervical Cancer	XGBoost	–	0.651	0.531 (81.6%)	0.504 (77.5%)	0.289 (44.4%)	0.432 (66.4%)
UCI Cervical Cancer	Cervical Cancer	MLP	–	0.427	0.452 (105.8%)	0.526 (123.3%)	0.461 (108.1%)	0.374 (87.7%)

Table 12: Time-series ablation: downstream utility (AUROC, Retention) with the clinical-validity layer *disabled*.

Dataset	Task	Classifier	Paper	Real (TRTR)	CoMed-CTGAN TSTR (Ret.)	CoMed-TVAE TSTR (Ret.)	CoMed-CG TSTR (Ret.)	CoMed-GC TSTR (Ret.)
MIMIC-III	Heart Failure Mortality	Logistic Regression	0.842	0.777	0.637 (82.0%)	0.863 (111.0%)	0.622 (80.0%)	0.824 (106.0%)
MIMIC-III	Heart Failure Mortality	Random Forest	–	0.858	0.488 (56.9%)	0.864 (100.8%)	0.461 (53.8%)	0.763 (89.0%)
MIMIC-III	Heart Failure Mortality	Gradient Boosting	–	0.802	0.365 (45.5%)	0.840 (104.7%)	0.517 (64.4%)	0.717 (89.4%)
MIMIC-III	Heart Failure Mortality	XGBoost	–	0.815	0.465 (57.0%)	0.868 (106.5%)	0.560 (68.8%)	0.719 (88.3%)
MIMIC-III	Heart Failure Mortality	MLP	–	0.472	0.536 (113.4%)	0.829 (175.6%)	0.452 (95.7%)	0.753 (159.4%)
MIMIC-III	Length-of-Stay	Logistic Regression	–	0.739	0.675 (91.3%)	0.710 (96.1%)	0.682 (92.3%)	0.696 (94.3%)
MIMIC-III	Length-of-Stay	Random Forest	–	0.765	0.691 (90.3%)	0.724 (94.6%)	0.696 (91.0%)	0.565 (73.9%)
MIMIC-III	Length-of-Stay	Gradient Boosting	–	0.772	0.701 (90.8%)	0.732 (94.7%)	0.698 (90.3%)	0.650 (84.2%)
MIMIC-III	Length-of-Stay	XGBoost	–	0.764	0.636 (83.2%)	0.746 (97.6%)	0.623 (81.5%)	0.517 (67.7%)
MIMIC-III	Length-of-Stay	MLP	–	0.746	0.648 (86.8%)	0.725 (97.2%)	0.644 (86.3%)	0.553 (74.0%)
MIMIC-IV	Heart Failure Mortality	Logistic Regression	–	0.821	0.766 (93.2%)	0.788 (96.0%)	0.770 (93.8%)	0.715 (87.0%)
MIMIC-IV	Heart Failure Mortality	Random Forest	–	0.829	0.758 (91.4%)	0.781 (94.2%)	0.766 (92.4%)	0.676 (81.5%)
MIMIC-IV	Heart Failure Mortality	Gradient Boosting	–	0.839	0.767 (91.3%)	0.792 (94.4%)	0.775 (92.3%)	0.625 (74.5%)
MIMIC-IV	Heart Failure Mortality	XGBoost	0.820	0.828	0.709 (85.7%)	0.808 (97.6%)	0.750 (90.6%)	0.566 (68.3%)
MIMIC-IV	Heart Failure Mortality	MLP	–	0.815	0.691 (84.8%)	0.787 (96.5%)	0.732 (89.8%)	0.467 (57.3%)
MIMIC-IV	Length-of-Stay	Logistic Regression	–	0.815	0.799 (98.1%)	0.741 (91.0%)	0.799 (98.1%)	0.779 (95.7%)
MIMIC-IV	Length-of-Stay	Random Forest	–	0.831	0.789 (95.0%)	0.752 (90.5%)	0.789 (95.0%)	0.740 (89.1%)
MIMIC-IV	Length-of-Stay	Gradient Boosting	–	0.829	0.806 (97.2%)	0.763 (92.0%)	0.804 (97.0%)	0.766 (92.3%)
MIMIC-IV	Length-of-Stay	XGBoost	0.860	0.835	0.773 (92.6%)	0.744 (89.1%)	0.776 (92.9%)	0.712 (85.3%)
MIMIC-IV	Length-of-Stay	MLP	–	0.831	0.792 (95.2%)	0.727 (87.4%)	0.790 (95.1%)	0.752 (90.5%)
eICU	Heart Failure Mortality	Logistic Regression	–	0.747	0.718 (96.1%)	0.748 (100.1%)	0.723 (96.7%)	0.726 (97.2%)
eICU	Heart Failure Mortality	Random Forest	–	0.753	0.668 (88.7%)	0.731 (97.0%)	0.694 (92.1%)	0.691 (91.7%)
eICU	Heart Failure Mortality	Gradient Boosting	–	0.750	0.663 (88.4%)	0.746 (99.5%)	0.705 (94.1%)	0.676 (90.2%)
eICU	Heart Failure Mortality	XGBoost	–	0.719	0.613 (85.3%)	0.753 (104.8%)	0.642 (89.4%)	0.611 (85.0%)
eICU	Heart Failure Mortality	MLP	–	0.724	0.677 (93.5%)	0.732 (101.1%)	0.656 (90.7%)	0.616 (85.1%)
eICU	Length-of-Stay	Logistic Regression	–	0.746	0.738 (98.9%)	0.703 (94.2%)	0.733 (98.2%)	0.711 (95.3%)
eICU	Length-of-Stay	Random Forest	–	0.762	0.724 (95.0%)	0.690 (90.6%)	0.730 (95.8%)	0.611 (80.1%)
eICU	Length-of-Stay	Gradient Boosting	–	0.763	0.746 (97.8%)	0.718 (94.1%)	0.743 (97.5%)	0.690 (90.5%)
eICU	Length-of-Stay	XGBoost	–	0.747	0.695 (93.0%)	0.709 (94.9%)	0.699 (93.6%)	0.566 (75.7%)
eICU	Length-of-Stay	MLP	–	0.757	0.734 (96.9%)	0.650 (85.8%)	0.730 (96.4%)	0.609 (80.4%)

7 Implementation Details: Hyperparameters

All synthetic data generation and downstream evaluation were run on Google Colab with a single NVIDIA L4 GPU.

Synthetic Data Generators

All four CoMed generators are built on the corresponding SDV single-table synthesizers. The following settings are shared across every dataset and generator: random seed= 42; GPU used when available (cuda); domain-validity constraints applied during fitting; enforce_min_max_values=True and enforce_rounding=False; categorical columns with at least 20 distinct values re-encoded with an rdt UniformEncoder before fitting; and one synthetic dataset generated at the same row count as the real training split.

- **CoMed-CTGAN** (CTGANSynthesizer): generator_lr=discriminator_lr= 2×10^{-4} ; generator_decay=discriminator_decay= 10^{-6} ; discriminator_steps= 1; pac= 10. Tabular: embedding_dim= 128, generator_dim=discriminator_dim= (256, 256). Time-series: embedding_dim= 512, generator_dim=discriminator_dim= (256, 512, 256).
- **CoMed-TVAE** (TVAESynthesizer): embedding_dim, compress_dims and decompress_dims mirror the CTGAN dimensions (tabular 128/(256, 256); time-series 512/(256, 512, 256)).
- **CoMed-CopulaGAN** (CopulaGANSynthesizer): embedding_dim, generator_dim, discriminator_dim, and pac= 10 as in CTGAN.
- **CoMed-GaussianCopula** (GaussianCopulaSynthesizer): statistical model with no epochs, batch size, or network dimensions; uses only the shared constraint and encoding settings above.

Training epochs and batch size scale with dataset size and are identical across CTGAN, TVAE, and CopulaGAN for a given dataset (GaussianCopula ignores them). For *tabular* data, total_epochs ranges from 300 (largest datasets, $\geq 100k$ rows) through 500–1200 (mid-size) to 1500 (small datasets), and batch_size ranges from 50 to 2000 (chosen even, divisible by pac= 10, and no larger than the row count). For *time-series* data, total_epochs is 1500 (2000 for the readmission tasks) and batch_size ranges from 500 to 5000, scaled to the number of rows.

Downstream Classifiers (Tabular)

- **Logistic Regression**: max_iter= 2000, class_weight=“balanced”, solver=“lbfgs”, random_state= 42.
- **Random Forest**: n_estimators= 300, class_weight=“balanced”, n_jobs= -1, random_state= 42.
- **Gradient Boosting**: n_estimators= 100 (60 for very large datasets), subsample= 1.0 (0.6 for very large datasets), max_depth= 3, random_state= 42.
- **XGBoost**: n_estimators= 300, max_depth= 6, learning_rate= 0.1, subsample= 0.9, colsample_bytree= 0.9, eval_metric=“logloss”, tree_method=“hist”, n_jobs= -1, random_state= 42.
- **MLP**: hidden_layer_sizes= (100,) (or (128, 64) for larger datasets), max_iter= 300 (120 for large datasets), early_stopping=True, n_iter_no_change= 15, random_state= 42.

Evaluation uses a stratified 80/20 train/test split with random_state= 42.

Downstream Classifiers (Time-Series)

- **Logistic Regression**: max_iter= 2000, class_weight=“balanced”, solver=“lbfgs”, random_state= 42.
- **Random Forest**: n_estimators= 400, class_weight=“balanced”, n_jobs= -1, random_state= 42.
- **Gradient Boosting**: random_state= 42 (defaults otherwise).
- **XGBoost**: n_estimators= 400, max_depth= 6, learning_rate= 0.1, subsample= 0.9, colsample_bytree= 0.9, eval_metric=“logloss”, scale_pos_weight computed from class counts, tree_method=“hist”, n_jobs= -1, random_state= 42.
- **MLP**: hidden_layer_sizes= (128, 64), max_iter= 500, early_stopping=True, n_iter_no_change= 15, validation_fraction= 0.1, random_state= 42.

The split and evaluation setup use random_seed= 42.